MAT242 - Topology

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Intro

The following are my notes for Topology which I will (hopefully) be taking in the fall of 2026. As these are my own personal notes I may skip ceirtain topics I already understand, be sloppy at times and this may cause trouble for anyone who wishes to use these notes for themselves so keep that in mind if you want to use these.

1 Topological spaces and continuity

We recall from Real Analysis that for a metric space (X, d) we have a notion of distance and can start describing open sets $U \subseteq X$ as sets where for every $x \in U$ there exists $\varepsilon > 0$ such that

$$B(x; \varepsilon) \subseteq U,$$

meaning that for any x in U there is some open ball around it contained entirely in U.

In Hilbert spaces, normed spaces, $\mathbb{R}^n$ (different types and instances of metric spaces), etc., this is the usual idea: an open set is one where every point has some wiggle room around it.

Topology starts from the observation that many arguments in Analysis don't really use the distance $d(x, y)$. They only use collections of open sets.

So instead of letting open sets be generated by some metric, why don't just specify which sets are open?

Intuitively, a topology on a set X is a choice about which sets are "open", whatever that may mean. Informally a topology tells us which points are "near" eachother without really specifing quantitatively what that may mean.

So we want the least structure possible to be able to talk about things like continuity, connectedness, compactness, convergence, etc.

Definition 1.1 (Topological space)**.** Let X be a set. A **topology** on X is a collection $\tau \subseteq \mathcal{P}(X)$ satisfying:

1. $\emptyset, X \in \tau$.
2. If $\{U_i\}_{i \in I}$ is any collection of elements in τ, then
$$\bigcup_{i \in I} U_i \in \tau.$$
3. If $U_1, \dots, U_n \in \tau$, then
$$U_1 \cap \cdots \cap U_n \in \tau.$$

The pair (X, τ) is called a **topological space**. The elements of τ are called **open sets**.

We will often refer to a topological space by just the set if it is clear what the topology on X is.

The axioms are not arbitrary, but abstracted from metric spaces. In metric spaces we already have these properties even if they are not explicitly required. For instance in $\mathbb{R}$,

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\}.$$

Each interval is open, but $\{0\}$ is not.

Intuitively, finite intersections preserve some positive amount of wiggle room while infinite intersections can squeeze it all away.

Examples of topological spaces

Example 1.1 ($\mathbb{R}$)**.** Let $X = \mathbb{R}$. The usual topology is the collection of subsets $U \subseteq \mathbb{R}$ such that for every $x \in U$, there exists $\varepsilon > 0$ with

$$(x - \varepsilon, x + \varepsilon) \subseteq U.$$

This is the topology we know from Real analysis. Examples of open sets include:

$$(0, 1),\ (-3, 7),\ \mathbb{R},\ \varnothing,$$

and also things like

$$(0, 1) \cup (2, 5) \cup (10, \infty).$$

Non-examples:

$$[0, 1],\ \mathbb{Q},\ \{0\}.$$

Note that $\mathbb{Q}$ can be made into a topological space by giving it the subspace topology from $\mathbb{R}$. More on that later.

Example 1.2 (Metric spaces)**.** Every metric space (X, d) gives a topology. Define τ_d by saying taht $U \subseteq X$ is open if for every $x \in U$ there exists $\varepsilon > 0$ such that

$$B(x; \varepsilon) \subseteq U.$$

Then τ_d is a topology on X. So every metric spaces is a topological space, but note that the opposite implication does not hold which is one of the reasons topology is more general.

Example 1.3 (The discrete topology)**.** Let X be any set. Let $\tau := \mathcal{P}(X)$. Then τ is a topology on X which we call the **discrete topology**.

Example 1.4 (The indiscrete topology)**.** Let X be any set. Define $\tau = \{\emptyset, X\}$. This is called the **indiscrete topology** or **trivial topology**.

Definition 1.2 (Closed set)**.** A subset $C \subseteq X$ is called **closed** if its complement is open:

$$C \text{ is closed} \Leftrightarrow X \setminus C \text{ is open.}$$

In a topological space, $\emptyset$ and X are closed. Arbitrary intersections of closed sets are closed and finite unions of closed sets are closed.

An important nuance is that closed does not mean not open. $\emptyset, X$ are both open and closed. In $\mathbb{R}$ (with the usual topology) we have that $\emptyset, \mathbb{R}$ are both open and closed while e.g. $[0, 1)$ is neither open nor closde.

Continuity

In Analysis, a function on metric spaces $f : X \to Y$ is continuous at $x \in X$ if small changes in x in $f(x) \in Y$. Recall the ε-δ definition:

$$\forall \varepsilon > 0,\ \exists \delta > 0 \text{ s.t. } d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Since we don't have access to metrics we must rewrite this.

Definition 1.3 (Continuity)**.** Let (X, τ_X) and (Y, τ_Y) be topological spaces. A function

$$f : X \to Y$$

is **continuous** if for every open set $V \in \tau_Y$, the preimage

$$f^{-1}(V)$$

is open in X.

Proposition 1.1. *Let (X, d_X) and (Y, d_Y) be metric spaces. A function*

$$f : X \to Y$$

is continuous in the ε-δ sense if and only if for every open set $V \subseteq Y$, the preimage

$$f^{-1}(V)$$

is open in X.

Proof. First suppose f is continuous in the ε-δ sense. Let $V \subseteq Y$ be open. We want to

show that $f^{-1}(V)$ is open in X. Take $x \in f^{-1}(V)$. Then $f(x) \in V$. Since V is open, there exists $\varepsilon > 0$ such that

$$B(f(x); \varepsilon) \subseteq V.$$

By continuity of f at x, there exists $\delta > 0$ such that

$$d_X(x, y) < \delta \Rightarrow d_Y(f(x), f(y)) < \varepsilon.$$

Thus, if $y \in B(x; \delta)$, then

$$f(y) \in B(f(x); \varepsilon) \subseteq V.$$

Therefore

$$y \in f^{-1}(V).$$

So

$$B(x; \delta) \subseteq f^{-1}(V).$$

Since every point of $f^{-1}(V)$ has a ball around it contained in $f^{-1}(V)$, the set $f^{-1}(V)$ is open.

Now suppose that for every open set $V \subseteq Y$, the preimage $f^{-1}(V)$ is open in X. Let $x \in X$, and let $\varepsilon > 0$. The ball

$$B(f(x); \delta)$$

is open in Y. Therefore

$$f^{-1}(B(f(x); \delta))$$

is open in Y. Since $x \in f^{-1}(B(f(x); \delta))$, there exists $\delta > 0$ such that

$$B(x; \delta) \subseteq f^{-1}(B(f(x); \delta)).$$

Thus, if

$$d_X(x, y) < \delta,$$

then

$$y \in f^{-1}(B(f(x); \delta)),$$

so

$$f(y) \in B(f(x); \delta).$$

Equivalently,

$$d_Y(f(x), f(y)) < \varepsilon.$$

Hence f is continuous at x. Since x was arbitrary, f is continuous. ■

Suppose τ_1 and τ_2 are two topologies on the same set X.

We say τ_1 is **finer** than τ_2 if

$$\tau_2 \subseteq \tau_1.$$

That means τ_1 has more open sets. Equivalently, τ_2 is **coarser** than τ_1. For every set X:

$$\{\emptyset, X\} \subseteq \tau \subseteq \mathcal{P}(X).$$

So the indiscrete topology is the coarsest topology possible, and the discrete topology is the finest topology possible.

This affects continuity. If the domain topology is finer, continuity is easier. If the target topology is finer, continuity is harder.

Example 1.5. Let X have the discrete topology and let Y be any topological space. Then every $f : X \to Y$ is continuous.
This is because $f^{-1}(V) \subseteq X$ and every subset of X is open.

Example 1.6. Let Y have the indiscrete topology. Then every function $f : X \to Y$ is continuous.
The preimages

$$f^{-1}(\varnothing) = \varnothing,\ f^{-1}(Y) = x$$

are both open in X.

Example 1.7. Let X be a set with two topologies τ_1, τ_2. Then $\mathrm{id} : (X, \tau_1) \to (X, \tau_2)$ is continuous exactly when $\tau_2 \subseteq \tau_1$.

First glimpse of homeomorphisms

Topology is about spaces up to continuous deformation. The correct notion of "same space" is not equality, but **homeomorphism**.

Definition 1.4 (Homeomorphism)**.** Let X, Y be topological spaces.
A function $X \to Y$ is a **homeomorphism** if

1. f is bijective,
2. f is continuous,
3. f^{-1} is continuous.

If such an f exists, then X and Y are homeomorphic.

Intuitively, homeomorphic spaces have the same topological shape. Think about those videos of coffee cups and donuts. This is what they meant.

Example 1.8. $(0, 1)$ and $\mathbb{R}$ are homeomorphic. One possible homeomorphism is

$$f : (0, 1) \to \mathbb{R},\ x \overset{f}{\mapsto} \tan\left(\pi x - \frac{\pi}{2}\right).$$

So $\mathbb{R}$ is homeomorphic to $(0, 1)$ even though $(0, 1)$ is "small" (bounded) in comparison to $\mathbb{R}$.

Lastly we remark that openness doesn't need to have anything to do with distance. In algebraic geometry and commutative algebra we look at the prime spectrum of a ring,

$$\text{Spec}(A)$$

which we endow with a topology called the **Zariski topology**. Closed sets are defined by ideals:

$$V(I) = \{\mathfrak{p} \in \text{Spec}(A) : I \subseteq \mathfrak{p}\}.$$

So the closed subsets are controlled by equations and ideals.

Exercises

Problem 1.1. Let X be any set. Prove that the discrete topology $\tau = \mathscr{P}(X)$ is a topology on X.

Proof. $\tau \subseteq \mathscr{P}(X)$ is ceirtainly satisfied. Furthermore $\emptyset, X \in \tau = \mathscr{P}(X)$ is also satisfied.

Now let I be some index set. Since $\mathscr{P}(X)$ contains all subsets of X and we necessarily have

$$\bigcup_{i \in I} U_i \subseteq X$$

for $U_i \in \mathscr{P}(X)$, and then we have

$$\bigcup_{i \in I} U_i \in \mathscr{P}(X) = \tau.$$

Thus τ is closed under arbitrary intersections.

Let $U_1, \dots, U_n \in \tau = \mathscr{P}(X)$. Once again since $U_i \subseteq X$ for all $1 \leq i \leq n$ we have

$$\bigcap_{i \in [n]} U_i \subseteq X$$

so

$$\bigcap_{i \in [n]} U_i \in \tau = \mathscr{P}(X).$$

■

Problem 1.2. Let X be any set. Prove that $\tau = \{\emptyset, X\}$ is a topology on X.

Proof. We have $\emptyset, X \in \tau$ by definition and also $\tau \subseteq \mathscr{P}(X)$.

Now notice that the only non-trivial collection of open sets in τ is the entireity of τ:

$$\emptyset, X,$$

and this collection is finite. Then $\emptyset \cup X = X \in \tau$ and $\emptyset \cap X = \emptyset \in \tau$.

Thus τ is closed under arbitrary unions and finite intersection adn hence forms a topology on X. ■

Problem 1.3. Let X be a topological space. Prove that arbitrary intersections of closed sets are closed.

Proof. Let I be some index set so that C_i is closed for every $i \in I$.

Then $X \setminus C_i$ is open for every $i \in I$. Then

$$\bigcup_{i \in I} X \setminus C_i = \bigcup_{i \in I} X \cap C_i^c$$

is open. Taking the complement we get

$$\left(\bigcup_{i \in I} X \cap C_i^c \right)^c = \bigcap_{i \in I} \emptyset \cup C_i = \bigcap_{i \in I} C_i$$

so

$$\bigcap_{i \in I} C_i$$

is closed. ■

Problem 1.4. Let τ_1, τ_2 be topologies on X. Prove that $\mathrm{id} : (X, \tau_1) \to (X, \tau_2)$ is continuous if and only if $\tau_2 \subseteq \tau_1$.

Proof. Suppose $\tau_2 \subseteq \tau_1$. Let $U \in \tau_2$. Then $U \in \tau_1$ and

$$\mathrm{id}^{-1}(U) = U \in \tau_1.$$

Thus the preimage of $U \in \tau_2$ is in τ_1. As U was arbitrary we have that id is continuous.

Suppose id is continuous. Then for any open set $U \in \tau_2$ we have that

$$\mathrm{id}^{-1}(U) = U \in \tau_1$$

hence

$$\tau_2 \subseteq \tau_1.$$

■

1.1 Some exercises from Section 13 of Munkres

Problem 1.5 (Problem 1). Let X be a topological space; let A be a subspace of X. Suppose that for each $x \in A$ there is an open set U containing x such that $U \subset A$. Show that A is open in X.

Proof. For each $x \in A$, let U_x denote the open set containing x such that $U_x \subset A$. We claim that

$$A = \bigcup_{x \in A} U_x.$$

For each $x \in A$, $U_x \subset A$ so

$$\bigcup_{x \in A} U_x \subseteq A.$$

Now, for some arbitrary $x \in A$, there exists an open set U_x such that $x \in U_x$. Thus

$$A \subseteq \bigcup_{x \in A} U_x.$$

The two inclusions give the result. ∎

Problem 1.6 (Problem 3). Show that the collection $\mathcal{T}_c$ given in Example 4 of §12 (of Munkres) is a topology on the set X.
Is the collection

$$\mathcal{T}_\infty = \{U \mid X \setminus U \text{ is infinite or empty or all of } X\}$$

a topology on X?
Note: For a set X, define $\mathcal{T}_c$ to be the collection of all subsets U of X such that $X \setminus U$ is either countable or all of X.

Proof. Since $X \setminus \emptyset = X$ so $\emptyset \in \mathcal{T}_c$. Furthermore, since $X - X = \emptyset$ which is finite and thus countable, we have $X \in \mathcal{T}_c$.

Let $\{U_\alpha\}$ be a family of open sets (ignoring the trivial case where $U_\alpha = \emptyset$ for every such U_α) in $\mathcal{T}_c$. By definition, for each U, we have that

$$|X \setminus U_\alpha| \leq \aleph_0.$$

Since

$$\left|\bigcup_\alpha U_\alpha\right| \geq |U_\alpha|$$

for any α, we have

$$\left|X \setminus \bigcup_\alpha U_\alpha\right| \leq |X \setminus U_\alpha| \leq \aleph_0.$$

Hence $\bigcup_\alpha U_\alpha \in \mathcal{T}_c$ so $\mathcal{T}_c$ is closed under arbitrary unions.

Now let $\{U_i\}_{i\in[n]}$ be a finite family of open sets. Assume also that $U_i \neq \emptyset$ for every i as that case is trivial. Then

$$X \setminus \bigcap_{i=1}^{n} U_i = \bigcup_{i=1}^{n} (X \setminus U_i).$$

By assumption, $X \setminus U_i$ is countable and since the union of countable sets is countable we have that $\mathcal{T}_c$ is closed under finite intersections.

Now to answer the second question, $\mathcal{T}_\infty$ is absolutely not a topology on X. Take for instance $X = \mathbb{N}$. Then $\{n\} \in \mathcal{T}_\infty$, but

$$\mathbb{N} \setminus \bigcup_{n\in\mathbb{N}\setminus\{7\}} \{n\}$$

is $\{7\}$ which is neither infinite, empty, or all of X. ■

Problem 1.7 (Problem 4)**.**

a) If $\{\mathcal{T}_\alpha\}$ is a family of topologies on X, show that $\bigcup \mathcal{T}_\alpha$ is a topology on X. Is $\bigcup \mathcal{T}_\alpha$ a topology on X?

Proof of (a). Suppose $\{\mathcal{T}_\alpha\}$ is a family of topologies on X. Then $\emptyset, X \in \mathcal{T}_\alpha$ for all α so

$$\emptyset, X \in \bigcap \mathcal{T}_\alpha.$$

Let $\{U_\beta\}$ be a family of open sets in $\bigcap \mathcal{T}_\alpha$. Then every U_β lies in $\mathcal{T}_\alpha$ for every α. Thus

$$\bigcup U_\beta \in \mathcal{T}_\alpha, \ \forall \alpha$$

and hence $\bigcup U_\beta$ lies in the intersection.

Now let $\{U_i\}_{i\in[n]}$ be a finite family of open sets in $\bigcap \mathcal{T}_\alpha$.

Again, $U_i \in \mathcal{T}_\alpha$ for every i and α and the argument is analogous to the one above.

The same does not hold for $\bigcup \mathcal{T}_\alpha$. Suppose $X = \{1, 2, 3\}$ with $\mathcal{T}_1 = \{\emptyset, X, 1\}$ and $\mathcal{T}_2 = \{\emptyset, X, 2\}$. Then $\{1\}$ and $\{2\}$ lie in their union, but $\{1, 2\}$ does not. ■

Problem 1.8 (Problem 6)**.** Show that the topologies $\mathbb{R}_\ell$ and $\mathbb{R}_K$ are not comparable.

Proof. Recall that the topology of $\mathbb{R}_\ell$ is

$$\mathcal{B}_\ell = \{[a, b) \mid a < b\},$$

and that the topology of $\mathbb{R}_K$ is

$$\mathscr{B}_K = \{(a,b) : a < b\} \cup \{(a,b) \setminus K : a < b\}$$

where

$$K = \left\{ \frac{1}{n} \,\middle|\, n \in \mathbb{Z}_+ \right\}.$$

These are not comparable. Take for instance $(-1,1) \setminus K \in \mathscr{B}_K$. If $\mathscr{B}_K \leq \mathscr{B}_\ell$ we should be able to find some $\varepsilon > 0$ so that $[0,\varepsilon) \subseteq (-1,1) \setminus K$, but for every $\varepsilon > 0$ we can choose n sufficiently large so that

$$\frac{1}{n} < \varepsilon.$$

Then $1/n \in [0,\varepsilon)$ implying $[0,\varepsilon) \not\subseteq (-1,1) \setminus K$. Thus $\mathscr{B}_K \not\subseteq \mathscr{B}_\ell$

For the other way consider $[0,1) \in \mathscr{B}_\ell$. Every K-neighbourhood at 0 is either (a,b) or $(a,b) \setminus K$ with $a < 0 < b$. Either way it contains some negative real numbers so no such basic neighborhood of 0 can fit in $[0,1)$. ■

Problem 1.9 (Problem 8)**.**

a) Apply Lemma 13.2 to show that the countable collection

$$\mathscr{B} = \{(a,b) \,|\, a < b,\ a,b \in \mathbb{Q}\}$$

is a basis that generates the standard topology on $\mathbb{R}$.

Proof of (a). Recall that Lemma 13.2 says that if X is a topological space, and $\mathscr{C}$ is a collection of open sets such that for each open set U of X and each $x \in U$ there is some element $C \in \mathscr{C}$ such that $x \in C \subset U$, then $\mathscr{C}$ is a basis for the topology on X.

Let U be open in the canonical topology on $\mathbb{R}$. Then $U = (a_0, b_0)$ for some $a_0, b_0 \in \mathbb{R}$ such that $a_0 < b_0$. By virtue of $\mathbb{Q}$ being dense in $\mathbb{R}$ we can, for each $\varepsilon > 0$ find some $a, b \in \mathbb{Q}$ suitably close:

$$a - a_0 < \varepsilon,\ b_0 - b < \varepsilon.$$

For any $x \in U = (a,b)$ let $\varepsilon = \min\{x - a_0, b_0 - x\} > 0$. Choose rational a and b (with $a < b$) so that $a - a_0, b_0 - b < \varepsilon$. Then

$$\begin{aligned} a - a_0 &< \varepsilon \\ &\leq x - a_0 \\ a &< x, \end{aligned}$$

and

$$\begin{aligned} b_0 - b &< \varepsilon \\ &\leq b_0 - x \\ -b &\leq -x \\ b &> x. \end{aligned}$$

Thus, by Lemma 13.2, $\mathscr{B}$ is a basis for the standard $\mathbb{R}$ topology.

This is a bit imprecise in places, but the idea is right. ■

2 Some problems to recap

Problem 2.1 (Munkres §16, Exercise 3)**.** Let

$$[-1, 1] \subseteq \mathbb{R}$$

with the subspace topology inherited from the standard topology on the reals. For each of the following sets, determine:

1. Is it open in Y?
2. Is it open in $\mathbb{R}$?

- $A = \left\{ x \;\middle|\; \frac{1}{2} < |x| < 1 \right\}$,
- $B = \left\{ x \;\middle|\; \frac{1}{2} < |x| \leq 1 \right\}$,
- $C = \left\{ x \;\middle|\; \frac{1}{2} \leq |x| < 1 \right\}$,
- $D = \left\{ x \;\middle|\; \frac{1}{2} \leq |x| \leq 1 \right\}$,

and

- $E = \{ x \mid 0 < |x| < 1 \text{ and } 1/x \notin \mathbb{Z}_+ \}$.

Sol. A set X is open in $\mathbb{R}$ if for every $x \in X$, there is some $\varepsilon > 0$ such that $(x-\varepsilon, x+\varepsilon) \subseteq X$. This is the same behavior inherited by Y.

We begin with A. We claim that this set is indeed open.

We have that

$$\frac{1}{2} < x < 1 \text{ or } -1 < x < -\frac{1}{2}.$$

Hence $A = (-1, -1/2) \cup (1/2, 1)$. The union of open sets is open hence A is open.

Now we do B. We claim that this set is not open. Let $x = 1$. There would have to exist $\varepsilon > 0$ so that $(x - \varepsilon, x + \varepsilon) \subseteq X$, but $x + \varepsilon > 1$ and hence not in B.

Now we do C. We also claim this set is not open using the same argument. An open ε-ball of $1/2$ would include $1 - \varepsilon/2 < 1/2$.

D fails for the same reason.

$$E = (-1, 0) \cup \bigcup_{n=1}^{\infty} \left(\frac{1}{n+1}, \frac{1}{n} \right),$$

a union of open sets, hence E is open. ■

Problem 2.2. Let

$$f : X \to Y$$

and suppose $\mathscr{B}$ is the basis for the topology on Y.
Prove that f is continuous if and only if

$$f^{-1}(B)$$

is open for every $B \in \mathscr{B}$.

Proof. Suppose f is continuous. Since $B \in \mathscr{B}$ are necessarily open in the topology they induce it is trivial that

$$f^{-1}(B)$$

is open, by definition.

Now suppose

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\}$$

is open for every $B \in \mathscr{B}$. Recall that every open set $U \subseteq Y$ can be written as

$$U = \bigcup_{\alpha} B_\alpha, \; B_\alpha \in \mathscr{B}.$$

Then

$$f^{-1}\left(\bigcup_{\alpha} B_\alpha \right) = \{x \in X \mid x \in f(B_\alpha) \text{ for at least one } \alpha\}.$$

In particular

$$f^{-1}\left(\bigcup_{\alpha} B_\alpha \right) = \bigcup_{\alpha} f^{-1}(B_\alpha)$$

and the preimage of B_α is, by assumption, open in X. As topologies are closed under arbitrary unions we get that $f^{-1}(U)$ is open in X. As U was arbitrary, this holds for any open $U \subseteq Y$.

■

Problem 2.3. Let X, Y, Z be topological spaces and let

$$f : Z \to X \times Y.$$

Write

$$f(z) = (f_1(z), f_2(z))$$

where f_1, f_2 are the projections to coordinates 1 and 2 respectively.
Prove that f is continuous if and only if f_1, f_2 are continuous.

Proof. The basis of the product topology is

$$\{U \times V \mid U \subseteq X \text{ is open, and } V \subseteq Y \text{ is open}\}.$$

Suppose f is continuous. Then

$$f_1 = \pi_1 \circ f, \text{ and } f_2 = \pi_2 \circ f$$

where π_1, π_2 are projections. Thus f_1, f_2 are compositions of continuous functions which is continuous.

Now suppose f_1, f_2 are continuous.

An open set $W \in Y$ can be written as

$$W = \bigcup_\alpha U_\alpha \times V_\alpha$$

Then $f^{-1}(W)$ is the set of all $z \in Z$ such that $f_1(z) \in U_\alpha$ for at least one α and $f_2(z) \in V_\alpha$ for at least one alpha so

$$f^{-1}(W) = \bigcup_\alpha f_1^{-1}(U_\alpha) \cap f_2^{-1}(V_\alpha).$$

As the $f_1^{-1}(U_\alpha), f_2^{-1}(V_\alpha)$ are open in Z, their finite union is open in Z and hence the union of $f_1^{-1}(U_\alpha), f_2^{-1}(V_\alpha)$ over α is open in Z. ∎

Problem 2.4. Let $\mathcal{T}, \mathcal{T}'$ be topologies on a set X with $\mathcal{T} \subseteq \mathcal{T}'$, meaning $\mathcal{T}$ is finer than $\mathcal{T}'$. Considering the identity map

$$\text{id} : X \to X$$

we can regard it in two ways, the one where $\mathcal{T}'$ is carried to $\mathcal{T}$ or the other way. For which of these is the identity map always continuous?

Proof. The map

$$\text{id} : (X, \mathcal{T}') \to (X, \mathcal{T})$$

is always continuous, the other doesn't need to be.

Let $U \in \mathcal{T}$ be open. Then

$$\mathrm{id}^{-1}(U) = U \in \mathcal{T}'.$$

Consider $\mathcal{T} = \{\emptyset, X\} \subseteq \mathcal{T}' = \{\emptyset, X, A\}$. Then the other identity map has

$$\mathrm{id}(A)^{-1} = A \notin \mathcal{T}.$$

■

3 Quotient spaces and quotient topology

Quotient topology will be in spirit antithetical to the product topology. The fundamental question is: if we glue points of some space X together, what topology should the space have?

We begin by recapping what equivalence relations are.

Definition 3.1. Let X be a set.
An equivalence relation, on X is relation $\sim$ satisfying reflexivity:

$$x \sim x$$

for every $x \in X$,
symmetry:

$$x \sim y \Leftrightarrow y \sim x,$$

and
transitivity:

$$x \sim y,\ y \sim z \Rightarrow x \sim z.$$

For $x \in X$ we denote the equivalence class

$$[x] = \{y \in X \mid y \sim x\}.$$

Equivalence classes partition X.

The quotient set is

$$X/\sim = \{[x] \mid x \in X\}.$$

There is a canonical map

$$q : X \to X/\sim,\ q(x) = [x].$$

This map is automatically surjective.

This is a nice construction in topology because these relations give us a sort of language for gluing.

Take $X = [0, 1]$. Suppose we identify two endpoints $0 \sim 1$, while every other endpoint is equivalent to only itself. Then the quotient

$$[0, 1]/\sim$$

is like a circle.

Likewise starting from a square

$$[0, 1] \times [0, 1]$$

and identifying the left and right edges appropriately gives a cylinder.

Now the natural question is what topology should be endowed on this space?

Suppose X is a topological space and let

$$q : X \to Y$$

be a surjection.

We want to make Y a topological space.

Since q tells us which points of X collapse onto the same point in Y, the natural idea is a set downstairs should be open exactly when its full preimage upstairs is open.

Definition 3.2 (Quotient topology)**.** The quotient topology on Y is

$$\tau_Y = \{U \subseteq Y \mid q^{-1}(U) \text{ open in } X\}.$$

Proposition 3.1. *The quotient topology is indeed a topology.*

Proof. First,

$$q^{-1}(\emptyset),\ q^{-1}(Y) = X.$$

Both are open in X.

Let $\{U_\alpha\}_\alpha \subseteq \tau_Y$. Then

$$q^{-1}\left(\bigcup_\alpha U_\alpha\right) = \bigcup_\alpha q^{-1}(U_\alpha).$$

Each $q^{-1}(U\alpha)$ is open in X so the union of these is open in X.

If $U_1, \dots, U_n \in \tau_Y$, than

$$q^{-1}\left(\bigcup_{i=1}^{n} U_i\right) = \bigcap_{i=1}^{n} q^{-1}(U_i)$$

which is the finite intersection of open sets in the domain and thus open. ■

We need the map to be surjective otherwise there could be points in the codomain which are never seen.

A surjective continuous map $q : X \to Y$ is called a quotient map if $U \subseteq Y$ if and only if $q^{-1}(U)$ is open in X.

A quotient space, on the other hand, is typically a space

$$X/\sim$$

equipped with the quotient topology induced by the canonical projection $q : X \to X/\sim$.

It is tempting to think that

$$U \subseteq X \text{ open} \Rightarrow q(U) \subseteq Y \text{ open.}$$

This is false in general.

The definition works with the preimage not with the image. Just like how topology in general behaves quite nicely with preimages.

Definition 3.3. A subset $A \subseteq X$ is called saturated with respect to q if it contains every fibre that it touches. Formally,

$$A = q^{-1}(q(A)).$$

In the case of an equivalence relation this means

$$\text{If } x \in A, \text{ then every point is equivalent to } x \text{ is also in } A.$$

Every set of the form $q^{-1}(A)$ is saturated.

Thus the quotient topology is really controlled by open saturated subsets of X.

Example 3.1. Let X be the closed unit interval. Define $0 \sim 1$ with no other nontrivial identifications.
Then $Y = X/\sim$ has one special point

$$[0] = [1].$$

Suppose $U \subseteq Y$ contains $[0]$. Its preimage must contain both 0 and 1. For $U$to be open in Y it must be open in the subspace $[0, 1]$. So it must contain something like

$$[0, \varepsilon) \cup (1 - \varepsilon, 1].$$

After quotienting, these two pieces become one neighborhood around the glued point.
That is exactly what neighborhoods around a point on the circle look like.

Now we arrive at the universal property.

Suppose

$$q : X \to Y$$

is a quotient map.

Let Z be another topological space and

$$f : X \to Y.$$

Then

$$f \text{ is continuous} \Leftrightarrow f \circ g \text{ is continuous.}$$

You can illustrate the difference between this- and the product topology's universal property particularly nice with diagrams in a category theory-esque way, but I'm lazy.

Now we pose a natural question.

Suppose

$$q : X \to X/\sim$$

is the quotient map. Imagine we have a continuous function

$$g : X \to Z.$$

When can it descend to a function $\overline{g} : X/\sim \to Z$ such that $g = \overline{g} \circ q$?

Set-theoretically, this is possible exactly when g is constant on equivalence classes:

$$x \sim y \Rightarrow g(x) = g(y).$$

If that happens, define

$$\overline{g}([x]) = g(x).$$

Because g is constant on equivalence classes, this is well-defined.

Then the quotient universal property gives

$$g = \overline{g} \circ q$$

continuous

$$\Rightarrow \overline{g} \text{ continuous.}$$

So quotient spaces let you convert functions on X that respect the identification into functions on the glued space.

This is perhaps the most useful operational idea:

If a continuous map cannot distinguish points you glue together,
it descends continuously to the quotient.

Problem 3.1. Let $X = [0,1]$ and define $0 \sim 1$ with all other equivalence classes singletons. Let

$$q : X \to X/\sim$$

be the quotient map. Consider

$$A = q([0,1/4) \cup (3/4,1]).$$

Try to determine whether A is open in the quotient space.

Sol. Let $B = [0,1/4) \cup (3/4,1]$. The forward inclusion

$$B \subseteq q^{-1}(q(B))$$

is always true. Indeed, if $x \in B$ then $q(x) \in q(B)$ so

$$x \in q^{-1}(q(B)).$$

For the reverse, suppose $x \in q^{-1}(q(B))$. Then

$$q(x) \in q(B),$$

so there exists $b \in B$ such that

$$q(x) = q(b).$$

But $q(x) = q(b)$ means that

$$x \sim b.$$

The only nontrivial class is $\{0,1\}$. Since Bcontains both $0,1$ whenever $b \in B$ every point is equivalent to b in B so $x \in B$.

The two inclusions mean that B is staurated under the equivalence relation. Hence A is open in the quotient space. ■

Problem 3.2.

a) Let $p : X \to Y$ be a continuous map. Show that if there is a continuous map $f : Y \to X$ such that $p \circ f$ equals the identity map of Y, then p is a quotient map.

b) If $A \subset X$, a retraction of X onto A is a continuous map $r : X \to A$ such that $r(a) = a$ for each $a \in A$. Show that a retraction is a quotient map.

Proof of (a). Recall that a quotient map $g : X \to Y$ is a surjection such that $U \subseteq Y$ is open if and only if $f^{-1}(U)$ is open.

Condition two is partially satisfied by p since, by assumption, $p^{-1}(U)$ is open when $U \subseteq Y$ is open.

Suppose p is not a surjection. Then there are $y \in Y$ such that, for every $x \in X$,

$$p(x) \neq y.$$

Then, $p \circ f \neq \mathrm{id}_Y$. $\mathrm{im}\, f \subseteq X$ so there is no $x \in \mathrm{im}\, f$ so that

$$p(x) = y.$$

In particular, $f(y) \in \mathrm{im}\, f \subseteq X$ so $f(y) \neq y$.

Thus we require that p is a surjection.

Now it remains to show that p carries open sets to open sets. Suppose $V \subseteq X$ is open. Then since f is continuous, $f^{-1}(V) \subseteq Y$ is open.

We have that

$$p \circ f(f^{-1}(V)) = f^{-1}(V)$$

so

$$p(V) = f^{-1}(V).$$

Thus p carries open sets to open sets.

Summing up, we have that p is a quotient map. ■

Proof of (b). Let $r : X \to A$ be a retraction, and let

$$i \hookrightarrow X$$

denote the inclusion map, defined by $i(a) = a$.

Since A has the subspace topology, i is continuous. Moreover, since r is a retraction,

$$(r \circ i)(a) = r(a) = a$$

for every $a \in A$. Hence

$$r \circ i = \mathrm{id}_A .$$

Thus r has a continuous right inverse. By part a, r is a quotient map. ■

Problem 3.3.

a) Define an equivalence relation on the plane $X = \mathbb{R}^2$ as follows:

$$x_0 \times y_0 \sim x_1 \times y_1 \text{ if } x_0 + y_0^2 = x_1 + y_1^2.$$

Let X^* be the corresponding quotient space. It is homeomorphic to a familiar space; what is it?

b) Repeat for the equivalence relation

$$(x_0, y_0) \sim (x_1, y_1) \text{ if } x_0^2 + y_0^2 = x_1^2 + y_1^2.$$

Solution to (a). Define $g([x, y]) : X^* \to \mathbb{R},\ g([x, y]) = x + y^2$.

Then,

$$(x_0, y_0) \sim (x_1, y_1) \Leftrightarrow g([x_0, y_0]) = g([x_1, y_1]).$$

Thus the equivalence classes are the fibres of g.

We claim this map is the continuous inverse of the quotient map $q : \mathbb{R}^2 \to \mathbb{R}$.

Well-definedness of the map is immediate.

It is also injective immediately from the definition.

For surjectivity, given $t \in \mathbb{R}$, simply take $(t, 0)$. Then $g([t, 0]) = t$.

Notice $x + y^2 = g \circ q$ is continuous. By the definining property of the quotient topology, g is continuous exactly when $g \circ q$ is. Thus g is continuous.

Define $s : \mathbb{R} \to \mathbb{R}^2$ by $s(t) = (t, 0)$. Then s is continuous.

Now $(q \circ s)(t) = [(t, 0)]$, but this is exactly the inverse of g. So g^{-1} is continuous.

Thus $X^* \cong \mathbb{R}$. ■

Solution sketch of (b). Proceed as above and define $g : \mathbb{R}^2 \to [0, \infty)$ by $g(x, y) = x^2 + y^2$ to get $X^* \cong [0, \infty)$. ■

3.1 Some more problems

Problem 3.4. Let $q : X \to Y$ be a surjection. Prove that for every $A \subseteq X$,

$$A \subseteq q^{-1}(q(A)).$$

Show that equality holds if and only if A is a union of fibers of q. In the equivalence-relation setting, formulate this as "A is a union of equivalence classes".

Proof. Let $A \subseteq X$. Then $q(A) \subseteq Y$. We look at the preimage

$$q^{-1}(q(A)) = \{x \in X \mid q(x) \in q(A)\}.$$

Clearly, this must include A. Otherwise there would be at least one $a \in A$ such that $q(a) \notin q(A)$, but $q(A) = \{q(a) \mid a \in A\}$ (Arguing about sets is ridiculous).

Suppose the preimage is exactly A, meaning A is saturated w.r.t q. Take any $a \in A$. Consider the fiber of q through a:

$$q^{-1}(\{q(a)\}).$$

We claim this entire fiber lies in A.

Indeed, if

$$x \in q^{-1}(\{q(a)\}),$$

then

$$q(x) = q(a).$$

Since $a \in A$,

$$q(a) \in q(A),$$

so $q(x) \in q(A)$.

Thus

$$x \in q^{-1}(q(A)).$$

By assumption, $q^{-1}(q(A)) = A$, so $x \in A$.

Conversely, suppose $A$is a union of fibers. Then it remains to prove that

$$q^{-1}(q(A)) \subseteq A.$$

Take $x \in q^{-1}(q(A))$. Then $q(x) \in q(A)$, so by definition of the image there is some $a \in A$ such that $q(x) = q(a)$. Thus x and a lie in the same fiber. Hence

$$q^{-1}(q(A)) \subseteq A.$$

■

Problem 3.5. Let $X \subseteq \mathbb{R}$ be the closed unit interval with $0 \sim 1$ and define $g : [0,1] \to \mathbb{R}^2$ as

$$g(t) = (\cos 2\pi t, \sin 2\pi t).$$

Show that g is constant on equivalence classes, so it induces a well-defined map

$$\overline{g} : X/\sim \to S^1.$$

Then use the quotient universal property to prove $\overline{g}$ is continuous.

Proof sketch. cos and sin are 2π-periodic so that

$$\cos(2\pi \cdot 0) = \cos(2\pi \cdot 1) \text{ and } \sin(2\pi \cdot 0) = \sin(2\pi \cdot 1).$$

This, together with the rest of the equivalence classes being singletons, makes g constant on all the equivalence classes. Thus $\overline{g} : X/\sim \to S^1$ is well-defined.

As this map is the composition of g (continuous) with the canonical projection $x \overset{\pi}{\mapsto} [x]$

then, by the universal property of quotient topology, $\overline{g}$ is continuous.

Furthermore $\overline{g}$ is invertible (I will not rigorously prove this) so

$$X/\sim \cong S_1$$

the 1-dimensional loop. ■

Problem 3.6. Let $X = \mathbb{R}$ and identify all integers to a single point, leaving all non-integers equivalent only to themselves. Give $X/\sim$ the quotient topology. Decide whether the quotient is Hausdorff.

Proof. For the quotient map $q : X \to X/\sim$ we have $A \subseteq X/\sim$ is open if and only its preimage is open in X.

The Hausdorff property is that for any two distinct points x, y we can find open neighborhoods for each which are disjoint.

We split it into two cases.

Case 1: $[\mathbb{Z}]$ and a noninteger point

Let $x \notin \mathbb{Z}$. Since $\mathbb{Z}$ is closed and discrete in $\mathbb{R}$, we can choose $\varepsilon > 0$ such that

$$[x - \varepsilon, x + \varepsilon] \cap \mathbb{Z} = \emptyset.$$

Put

$$V = (x - \varepsilon/2, x + \varepsilon/2)$$

and

$$U = R/[x - \varepsilon, x + \varepsilon].$$

Then U and V are disjoint w.r.t q. Therefore

$$q^{-1}(q(U)) = U,\ q^{-1}(q(V)) = V.$$

Since $U, V$are open, the quotient topology implies that $Q(U), Q(V)$are open in $\mathbb{R}/\sim$. They are also disjoint, because U, V are disjoint saturated sets. Finally

$$[\mathbb{Z}] \in q(U)$$

and

$$[x] \in q(V).$$

Thus $[\mathbb{Z}], [x]$ have disjoint open neighborhoods.

The case for two nonintegers is easy from the typical topology on $\mathbb{R}$. ■

4 Connectedness, components, compactness

From §23 of Munkres recall that a space X is connected if we cannot write $X = U \sqcup V$ where U, V are disjoint, nonempty, open subsets of X.

Equivalently, X is connected if and only if the only clopen subsets of X are X and $\emptyset$.

Also, X connected and $f : X \to Y$ continuous implies $f(X)$ connected.

4.1 §24, Connected subsets of the real line

The main theorem we will be keeping in mind is $A \subseteq \mathbb{R}$ is connected if and only if A is an interval, which is perhaps unsurprising.

Interval should be understood in the broad order-theoretic sense, meaning

$$x, z \in A,\ x < y < z \Rightarrow y \in A.$$

Such sets are also called convex subsets of the ordered set $\mathbb{R}$ in some contexts. Munkres develops this more generally for linear continua later, I believe.

The claim now is that connected subsets must be intervals.

Suppose $A \subseteq \mathbb{R}$ is connected. Take $x, z \in A,\ x < z$. Suppose some y satisfies $x < y < z$, but does not belong to A.

Then

$$A \cap (-\infty, y)$$

and

$$A \cap (y, \infty)$$

are disjoint nonempty open subsets of A, and together they cover A.

They are nonempty because x is in the first and z is in the second.

Hence they give a separation of A, contradicting connectedness. Therefore every point between x and z lies in A. So A is an interval.

4.2 The intermediate Value Theorem in topology

Now suppose X is connected and $f : X \to \mathbb{R}$ is continuous. Because continuous images of connected spaces are connected we have that $f(X)$ is connected, but connected subsets of $\mathbb{R}$ are intervals.

Therefore if

$$f(a) < r < f(b),$$

then

$$r \in f(X).$$

So there exists $c \in X$ with $f(x) = r$.

4.3 Path connectedness

Path connectedness is a strictly stronger notion that connectedness.

A path from x to y in X is a continuous map

$$\gamma : [0,1] \to X$$

such that

$$\gamma(0) = x,\ \gamma(1) = y.$$

A space X is path connected if every pair $x, y \in X$ can be joined by a path. Intuitively, connected says we cannot split the space, while path connected means we can travel continuously beetween all distinct points.

Proposition 4.1. *Path connectedness implies connectedness.*

Proof. Suppose X is path connected, but disconnected. Then $X = U \cup V$ is a separation. Choose $x \in U$ and $y \in V$. There is a path $\gamma : [0,1] \to X$ from x to y, but $[0,1]$ is connected, so its continuous image $\gamma([0,1])$ is connected. Yet this image intersects both U and V, which would separate it, a contradiction. ∎

4.4 §25, Components: how a disconnected space splits

Connectedness is a yes/no question asking "Is the whole space connected?". Components are slightly more refined and instead ask "What are the largest connected chunks of the space?".

For $x \in X$ define the components of x to be the union of all connected subsets of X containing x.

Equivalently, define $x \sim y$ if there exists a connected subset of X containing both x and y. The equivalence classes are the components of X. So a component is a maximal connected subspace.

Suppose

$$C_x = \bigcup_\alpha A_\alpha,$$

where each A_α is connected and contains x. Every A_α has the common point x.

The union of connected sets having a common point is connected.

Therefore C_x is connected. And since it contains every connected set containing x, it is maximal.

Furthermore, components partition the space.

Every point lies in its own component. Two components are either disjoint or equal. Suppose C_1, C_2 intersect. Then $C_1 \cup C_2$ is connected because the two connected sets have a common point. However, both C_1, C_2 are maximal connected sets so $C_1 = C_2$.

Thus X is partitioned into connected pieces.

4.5 Components are closed

This is an important theorem: Every component of a space is closed.

The main ingredient in this is an important one, namely that A connected $\Rightarrow \overline{A}$ connected. If C is a component, then its closure $\overline{C}$ is connected, but $C \subseteq \overline{C}$ and C was maximal so $\overline{C} = C$. Hence it is closed.

4.6 Path components

Similarly, define $x \sim_p y$ if there exists a path joining x and y. The equivalence classes are called path components. Every path component lies inside a component because path connected $\Rightarrow$ connected.

So conceptually, path components $\subseteq$ components and they can be strictly smaller.

4.7 Local connectedness

A space can faily to be connected globally, but look connected locally.

A space X is locally connected at x if every neighborhood U of x contains a connected neighborhood V of x.

That is,

$$x \in V \subseteq U$$

with V connected and a neighborhood of x.

It is locally connected if this holds at every $x \in X$. Munkres gives an analogous definition for local path connectedness. For example, $\mathbb{R} \setminus \{0\}$ is not connected, but it is locally connected, meaning around every point you can choose a sufficiently small interval not crossing 0.

4.8 Local theorem

Munkres gives the useful characterization,

$$X \text{ is locally connected}$$

$$\Updownarrow$$

$$\text{every component of every open subset of } X \text{ is open in } X.$$

There is, of course, an analogous statement with path components and locally path connected spaces.

Now, why should this make sense?

Suppose X is locally connected, $U \subseteq X$ is open, and C is a component of U. Take $x \in C$.

Because U is a neighborhood of x, local connectednes gives a connected neighborhood V_x with $x \in V_x \subseteq U$.

Since V_x is connected and intersects C, maximality of C gives $V_x \subseteq C$. Do this for every $x \in C$. Then

$$C = \bigcup_{x \in C} V_x.$$

Hence C is open.

Also, X locally path connected $\Rightarrow$ components $=$ path components.

4.9 §26, Compactness

Connectedness asks whether the space can be pulled apart, while compactness asks whether infinitely much local information can be reduced to finitely much information. Let $\mathcal{U} = \{U_\alpha\}_{\alpha \in A}$ be a collectioon of open sets in X.

It is an open cover of X if

$$X \subseteq \bigcup_{\alpha \in A} U_\alpha.$$

The space X is compact if every open cover has a finite subcover, meaning

$$X \subseteq \bigcup_{\alpha \in A} U_\alpha \Rightarrow X \subseteq U_{\alpha_1} \cup \cdots \cup U_{\alpha_n}$$

for some finite choice of indices.

As a first example, recall from analysis that $[0, 1]$ is compact, but $(0, 1)$ is not. For example, $U_n = (1/n, 1)$ cannot be reduced to a finite cover.

4.10 Closed subsets of compact spaces

Theorem 4.1. *X is compact and $A \subseteq X$ closed implies that A is compact.*

Proof. Let $\{U_\alpha\}$ be an open cover of A. Because A is closed, $X \setminus A$ is open. Then $\{U_\alpha\} \cup \{X \setminus A\}$ covers X and can thus be reduced to a finite cover. Discard $X \setminus A$ if it was chosen. The remaining finitely many U_α's cover A. ■

4.11 Compact subsets of Hausdorff spaces are closed

The converse is generally not true.

In a Hausdorff space, $A \subseteq X$, A compact, X Hausdorff $\Rightarrow$ A closed.

Take $x \in X \setminus A$. For every $a \in A$, Hausdorffness gives disjoint open $U_a \ni a$ and $V_a \ni x$.

The sets U_a cover A. Compactness gives finitely many $U_{a_1}, \dots, U_{a_n}$.

Let

$$V = V_{a_1} \cap \cdots \cap V_{a_n}.$$

This is open, contains x, and is disjoint from $U_{a_1} \cup \cdots \cup U_{a_n}$, which contains A. Thus $V \cap A = \emptyset$. So every point outside A has an open neighborhood not intersecting with A. Therefore $X \setminus A$ is open, hence A is closed.

4.12 Two nice theorems

Another nice theorem I'll state informally:

$$X \text{ compact, } f : X \to Y \text{ continuous} \Rightarrow f(X) \text{ compact.}$$

Let $\{U_\alpha\}$ be an open cover of $f(X)$. Then $\{f^{-1}(U_\alpha)\}$ covers X. Compactness lets us reduce to finitely many and therefore we can find a finite cover of the image od f.

Theorem 4.2. *Suppose $f : X \to Y$ is continuous and bijective with X compact and Y Hausdorff. Then f is a homeomorphism.*

Proof sketch. Take a closed set $C \subseteq X$. Since X is compact, C is compact. Continuous images preserve that property so $f(C)$ is compact in Y. Since Y is Hausdorff, $f(C)$ is closed. Thus f maps closed sets to closed sets. Since f is a bijection, this implies f^{-1} is continuous. ∎

4.13 Some problems

Problem 4.1 (Munkres 24.1)**.**

a) Show that no two of the spaces $(0,1)$, $(0,1]$, $[0,1]$ are homeomorpnic.

b) Suppose there exist imbedddings $f : X \to Y$ and $g : Y \to X$. Show by means of an example that X, Y need not be homeomorphic.

c) Show $\mathbb{R}^n$ and $\mathbb{R}$ are not homeomorphic if $n > 1$.

Proof of (a). Notice that $[0,1]$ is compact while $(0,1)$ is not. The latter is not since, if we take for instance the cover

$$\{(1/n, 1)\}_{n \in \mathbb{N}}$$

of $(0,1)$ this cannot be reduced to a finite cover since for every n we can find some $m > n$ such that

$$0 < \underbrace{1/m}_{\in(0,1)} < 1/n.$$

So $[0,1]$ and $(0,1)$ cannot be homeomorphic since continuous images would preserve

compactness.

$[0,1]$ and $(0,1]$ cannot be homeomorphic for the same reason.

Lastly $(0,1)$ is not homeomorphic to $(0,1]$. Suppose they were. Let $f : (0,1] \to (0,1)$ be the corresponding homeomorphism. Then

$$f|_{(0,1]\setminus\{1\}} : (0,1) \to (0,1) \setminus f(\{1\})$$

would be a homeomorphism. Let $x = f(1)$. For any point in $y \in (0,1)$, its removal leaves

$$(0,1) \setminus \{y\} = (0,y) \cup (y,1)$$

and in particular

$$(0,1) \setminus f(\{1\}) = (0, f(1)) \cup (f(1), 1)$$

so the space is disconnected, while $(0,1] \setminus \{1\} = (0,1)$ is connected, a contradiction. ■

Problem 4.2 (Munkres 25.4)**.** Let X be locally path connected. Show that every connected open set in X is path connected.

Proof. Let $U \subseteq X$ be open and connected. We show that U is path connected.

Fix $x \in U$ and let

$$P_x = \{z \in U \ : \text{there exists a path in } U \text{ from } x \text{ to } z\}.$$

Thus P_x is the path component of x in U.

We first show that P_x is open in U. Let $z \in P_x$. Since U is an open neighborhood of z and X is locally path connected, there exists an open path-connected set V_z such that

$$z \in V_z \subseteq U.$$

For every $w \in V_z$, there is a path from z to w inside V_z. Since $z \in P_x$, there is also a path from x to z inside U. Composing these paths gives a path from x to w, so $w \in P_x$. Hence

$$V_z \subseteq P_x,$$

and therefore P_x is open.

We similarly show that $U \setminus P_x$ is open. Let $z \in U \setminus P_x$. Again choose an open path-connected neighborhood

$$z \in V_z \subseteq U.$$

Suppose that $V_z \cap P_x \neq \emptyset$, and choose $w \in V_z \cap P_x$. There is a path from x to w, and since V_z is path connected, there is a path from w to z. Composing them would give a path from x to z, contradicting $z \notin P_x$. Hence

$$V_z \subseteq U \setminus P_x.$$

Thus $U \setminus P_x$ is also open.

Therefore P_x is both open and closed in U. Since U is connected and P_x is nonempty, we must have

$$P_x = U.$$

Hence every point of U can be joined to x by a path, so U is path connected.

■

Problem 4.3. Let X be a topological space, and let $A, B \subseteq X$ be connected subspaces such that $A \cap B \neq \emptyset$. Prove that $A \cup B$ is connected.

Proof. We are free to discard the trivial case where $A = B$.

Since A, B are connected, the only continuous functions $f : A \to \{0,1\}$ and $g : B \to \{0,1\}$ are constant functions.

Suppose $A \cup B$ is not connected, meaning we can find a continuous $h : A \cup B \to \{0,1\}$ which is non-constant.

Necessarily,

$$h\,\big|_A = c_A \in \{0,1\}$$

is constant, and

$$h\,\big|_B = c_B \in \{0,1\}$$

is constant, since they are connected spaces.

We assume h is non-constant. So there exists some $x, y \in A \cup B$ such that

$$h(x) \neq h(y).$$

Then, necessarily, these points cannot be in the intersection of A and B. So one of these points are in $A \setminus B$ or $B \setminus A$. Without loss of generality, suppose $x \in A \setminus B$. Then we require $y \in B \setminus A$ since otherwise we would have

$$c_A = h(x) \neq h(y) = c_A.$$

From this we deduce $c_A \neq c_B$, but then, for $z \in A \cap B$, we would have

$$h(z) = h\,\big|_A\,(z) = h\,\big|_B\,(z)$$

so

$$c_A = c_B,$$

a contradiction. ■

Problem 4.4. Let $\{A_\alpha\}_{\alpha \in I}$ be a collection of connected subspaces of X. Suppose there exists a point $p \in X$ such that $p \in A_\alpha$ for every $\alpha \in I$.

Prove that

$$\bigcup_{\alpha \in I} A_\alpha$$

is connected.

Proof. We suppose A_α are all connected, so for any continuous $f_\alpha : A_\alpha \to \{0,1\}$, the function must be constant.

Let $\varphi : \bigcup_\alpha A_\alpha \to \{0,1\}$ be continuous.

Then, necessarily,

$$\varphi\big|_{A_\alpha} : A_\alpha \to \{0,1\}$$

is constant (also the restriction remains continuous).

We have

$$\varphi(p) = c \in \{0,1\}$$

and in particular, since $p \in A_\alpha$ for every $\alpha \in I$, and the restriction to A_α is constant, we have

$$\operatorname{im} \varphi\big|_{A_\alpha} = \{c\}$$

for every $\alpha \in I$. Thus

$$\operatorname{im} \varphi = \{c\}$$

is constant, and hence

$$\bigcup_{\alpha} A_\alpha$$

is connected. ∎

Problem 4.5. Let $A \subseteq X$ be connected. Prove that its closure $\overline{A}$ is connected.

Proof. Suppose A is a connected subspace of some space X. Then any continuous function $f : A \to \{0,1\}$ must be constant.

Recall that $\overline{A}$ is the smallest closed set containing A, equivalently it is the intersection of all closed subsets of X containing A. Denote $f(A) = \{c_A\}$.

Let $f : \overline{A} \to \{0,1\}$ be some continuous function.

Suppose, $x, y \in \overline{A}$ with

$$f(x) = c_x,\ f(y) = c_y.$$

The fiber of c_x is an open set V_x of $\overline{A}$ containing x. Likewise, the fiber of c_y is an open set V_y of $\overline{A}$ containing y.

Then $f(V_x \cap A) = \{c_x\}$ and $f(V_y \cap A) = \{c_y\}$, but since these are subsets of A, and

$$\operatorname{im} f\big|_A = \{c_A\},$$

we must have

$$c_x = c_A = c_y.$$

Therefore the space is connected.

Note: I should have been more rigorous in explaining that we can do this by arguing that since the points were in the closure, any open neighborhood necessarily intersects A.

■

Problem 4.6. Let C be a connected component of a topological space X. Prove that C is closed in X.
You may use A connected $\Rightarrow \overline{A}$ connected.

Proof. C is a maximal connected subset.

Thus there does not exist any $D \subseteq X$ containing C which is also connected.

In particular, since $C \subseteq \overline{C}$ we must have that

$$C = \overline{C}.$$

Otherwise, $\overline{C}$ would be a larger connected subspace (C connected $\Rightarrow \overline{C}$ connected) containing all of C, contradicting the maximality of C. ■

Problem 4.7. Let X be locally connected. Let $U \subseteq X$ be open, and let C be a connected component of U.
Prove that C is open in X.

Proof. Since X is locally connected, then for every $x \in X$ we can, for every open neighborhood $U_x \supseteq x$, find an open connected set $V_x \supseteq x$ contained in U_x.

We suppose U is an open set in X and that C is a maximal connected subset of U.

For any $x \in C$, we can find a connected open neighborhood V_x with $V_x \subseteq U$.

Since C is maximal, $x \in C$, and V_x is connected we have $V_x \subseteq C$ for every x (might need one more step to be completely justified). Then

$$\bigcup_x V_x = C.$$

Since open sets are closed under arbitrary unions, C is open.

■

5 §27 and 29

In $\mathbb{R}$, we can study compactness in a more familiar setting (more Real Analysis, damn it).

We recall, from Real Analysis, that in $\mathbb{R}$, compactness corresponds to being closed and bounded.

However, this equivalence is special to Euclidian-like spaces. We have seen that this does not generally hold.

5.1 Closed intervals are compact

The first major result is that

$$[a, b] \subseteq \mathbb{R}$$

is compact, always.

Essentially, this is just Heine-Borel in one dimension. Munkres uses the least upper bound property of $\mathbb{R}$.

Suppose $\mathcal{U}$ is some open cover of $[a, b]$.

Define

$$C = \{x \in [a, b] \mid [a, x] \text{ can be covered by finitely many members of } \mathcal{U}\}.$$

Tge strategy is to show that this set is nonempty, let $c = \sup C$, and prove that $c = b$.

Suppose instead $c < b$. Since $\mathcal{U}$ covers $[a, b]$, some $U \in \mathcal{U}$ contains c.

Because U is open it contains an interval $(c - \varepsilon, c + \varepsilon)$.

Since $c = \sup C$, there is some $x \in C$ with $c - \varepsilon < x \leq c$.

Now $[a, b]$ already has a finite subcover, and U extends that finite cover beyond c.

So some point larger than c belongs to C, a contradiction. So we conclude $c = b$, so the interval has a finite subcover.

5.2 Heine-Borel (again)

From the previous result, together with the general compactness results from §26, we get that

$$A \subseteq \mathbb{R}^n \text{ is compact} \Leftrightarrow A \text{ is closed and bounded.}$$

Proof sketch. $\Rightarrow$.

Since $\mathbb{R}^n$ is Hausdorff, A compact implies that A is closed.

For unboundedness, suppose A were unbounded. Then

$$\{B(0, n) \mid n \in \mathbb{N}\}$$

is an open cover of A.

Any finite subcollection is contained in one largest ball

$$B(0, N),$$

which cannot cover unbounded A.

Contradiction, meaning A is bounded.

$\Leftarrow$.

If $A \subseteq \mathbb{R}^n$ is bounded, then for some M,

$$A \subseteq [-M, M]^n.$$

The box is compact. Since A is closed in $\mathbb{R}^n$ it is closed and compact in the box. Closed subsets of compact spaces are compact so compactnjess of A follows. ■

5.3 Extreme Value Theorem

Suppose X is compact and $f : X \to \mathbb{R}$ is continuous.

Then $f(X)$ has a maximum and minimum. So there exists $x_{\text{min}}, x_{\text{max}} \in X$ such that $f(x_{\text{min}}) \leq f(x) \leq f(x_{\text{max}})$ for all $x \in X$.

So what we learned in calculus,

continuous function on closed bounded interval attains max/min

is generalized to

continous images preserve compactness.

5.4 Uniform continuity

Recall that continuity in Real Analysis meant

$$\forall x,\ \forall \varepsilon > 0,\ \exists \delta_x > 0 : d(x, y) < \delta_x \Rightarrow d'(f(x), f(y)) < \varepsilon$$

with the distinction that δ_x may depend on x.

Recall that uniform continuity requires that for every $\varepsilon > 0$ we can find some point-independent $\delta > 0$ such that the condition holds.

> **Theorem 5.1.** *If X is compact and $f : X \to Y$ is continuous between metric spaces, then f is uniformly continuous.*

Intuitively this makes sense, since the compactness allows us to have finitely many local choices of δ for which we can simply take the minimum.

5.5 Distance from a point to a set

We define $d(x, A) = \inf\{d(x, a) \mid a \in A\}$.

If A is compact, the infinmum is achieved, meaning there is actually some $a_0 \in A$ such that $d(x, A) = d(x, a_0)$.

With the previous fact this is trivial to prove, as $a \mapsto d(x, a)$ is continuous on compact A, hence attains its minimum.

5.6 Closed and bounded does not imply compact in general

In $\mathbb{R}^n$, compact coincides precisely with being closed and bounded, but in arbitrary metric spaces this fails.

For instance, an infinite set with the discrete metric $d(x, y) = 0$ whenver $x = y$ and 1 otherwise, is nicely bounded and the whole space is closed. However, the singleton cover $\{\{x\} \mid x \in X\}$ has no finite subcover, meaning X is not compact.

6 §29 Local compactness

Definition 6.1. A space X is locally compact at x if there exists a compact subspace $C \subseteq X$ that contains a neighborhood of x.
Equivalently, there exists an open neighborhood U of x such that $x \in U \subseteq C$ for some compact C.
If it holds for every $x \in X$ we say the space X is locally compact.

Take for example $\mathbb{R}$. This space is not compact, but it is obviously locally compact since every point x lies in

$$(x - 1, x + 1) \subseteq [x - 1, x + 1],$$

and the closed interval is compact.

Similarly, $\mathbb{R}^n$ is not compact, though it is locally compact.

6.1 Compact implies locally compact

Unsurprisingly, compactness of X implies that every point $x \in X$ has the neighborhood X contained in the compact set X. Not terribly complicated.

The converse fails. Also not very surprising.

6.2 $\mathbb{Q}$ is not locally compact

In a useful contrast with the reals, the rationals is not locally compact.

Take $x \in \mathbb{Q}$.

Any neighborhood in $\mathbb{Q}$ contains somethign like $(a, b) \cap \mathbb{Q}$. Its closure is roughly $[a, b] \cap \mathbb{Q}$, but this is not compact.

One way to see why is that there are rational sequences convering in $\mathbb{R}$ to irrational numbers.

For instance, $q_n \to \sqrt{2}$ with all $q_n \in \mathbb{Q}$.

Suhc a sequence has no convergent subsequence in the rationals with a limit in the rationals. So $\mathbb{Q}$ has no compact neighborhood around a point.

6.3 Stronger local form in Hausdorff spaces

Munkres proves that if X is Hausdorff, then local compactness is equivalent to the following more natural statement:

For every $x \in X$ and every neighborhood U of x, there exists a neighborhood V of x such that $\overline{V}$ is compact, and

$$\overline{V} \subseteq U.$$

As per usual, Hausdorff spaces are nicely behaved.

In particular, we can find arbitrarily small relatively compact neighborhoods.

6.4 One-point compactification

Suppose X is locally compact Hausdorff, but not compact.

Add a new point

$$X^* = X \cup \{\infty\}.$$

How should neighborhood of ∞ look?

The idea will be that being close to ∞ means being outside of some compact part of X.

So define neighborhood of ∞ by sets of the form $X^* \setminus K$, where $K \subseteq X$ is compact.

Together with the original open sets of X, these generate a topology on X^*.

Then X^* is compact Hausdorff.

This is called the one-point compactification of X.

It sounds weird, but it makes sense (trust me bro).

Suppose we have an open cover of X^*. Some open set U contains ∞. By definition, $X^* \setminus U$ is a compact subset $K \subseteq X$.

So U already covers everything except a compact chunk.

The rest of the open cover covers K.

Compactness of K gives finitely many sets covering it.

Add U. Then we are done.

6.5 Some examples and then practice problems

The one-point compactification of $\mathbb{R}$ is homeomorphic to S^1.

Likewise, $\mathbb{R}^2 \cup \{\infty\}$ is homeomorphic to S^2 and more generally,

$$(\mathbb{R}^n)^* \cong S^n.$$

Problem 6.1. Let X be a compact space and let $f : X \to \mathbb{R}$ be continuous. Prove that f attains a maximum and minimum.
Prove it using only the fact that continuious images of compact spaces are compact and what compact subsets of $\mathbb{R}$ look like.

Proof. We let X be compact and $f : X \to \mathbb{R}$ be continuous.

Then

$$f(X)$$

is compact. By the Heine-Borel theorem, $f(X)$ is closed and bounded.

Then for any x in the fiber of $\arg\max f$, $f(x) \geq f(y)$ for every $y \in X$ and similarly, any x in the fiber of $\arg\min f$ satisfies $f(x) \leq f(y)$ for all $y \in X$.

Importantly, $\arg\max f = \sup f(X)$ which is finite (from boundedness) and in $f(X)$ since $f(X)$ is closed. Analogously, $\arg\min f = \inf f(x)$ which, once again, is finite and in $f(X)$. ■

Problem 6.2. Let X be a noncompact locally compact Hausdorff space. Define $X^* = X \cup \{\infty\}$, and declare a set $U \subseteq X^*$ open if either:

- $U \subseteq X$ and U is open in X, or
- $\infty \in U$ and $X^* \setminus U$ is compact in X.

Prove that X^*, the one-point compactification of X, is compact.

Proof. Let $U \subseteq X^*$ be some open set.

Then, recall throughout that we either have that U is open in X or $\infty \in U$ and $X^* \setminus U$ is compact in X.

Let $\mathcal{U} = \{U_\alpha\}_\alpha$ be some cover of X^*,

$$\bigcup_\alpha U_\alpha = X^*.$$

Necessarily, one of these U_α must contain ∞. Then

$$X^* \setminus U_\alpha$$

is compact, meaning $\mathcal{U}$, as a cover of $X^* \setminus U_\alpha$ can be reduced to a finite cover

$$\bigcup_{i=1}^{n} U_i = X^* \setminus U_\alpha.$$

Then,

$$U_\alpha \cup \bigcup_{i=1}^{n} U_i = U_\alpha \cup X^* \setminus U_\alpha = X^*.$$

So we have found a finite cover from the general cover, meaning X^* is compact. ■

Problem 6.3. Let X be a noncompact locally compact Hausdorff space, and let

$$X^* = X \cup \{\infty\}$$

have the one-point compactification topology.
Prove that X^* is Hausdorff.

Proof. X is already Hausdorff so separating two points $x, y \in X$ is not an issue.

Fix $x \in X^* \setminus \{\infty\}$. We will look at separating x and ∞.

A set $U \subseteq X^*$ containing ∞ is open if and only if $X^* \setminus U$ is compact in X.

Ideally we would want to prove that there must exist some open set U with $x \notin U$ for our arbitrary $x \in X = X^* \setminus \{\infty\}$, for which $X^* \setminus U$ is compact.

Since X is locally compact (and Hausdorff), we can, for any open neighborhood V of x find $W \subseteq V$ containing x such that $\overline{W}$ is compact.

Since X is Hausdorff and $\overline{W}$ is compact, we have that $\overline{W}$ is closed.

Thus $X \setminus \overline{W}$ is open in X. Therefore, $X^* \setminus \overline{W}$ is open in X^*. We don't really need to make this observation, since it suffices that $\overline{W}$ is compact $\Rightarrow X^* \setminus \overline{W}$ is open in X^*.

Furthermore, $\infty \in X^* \setminus \overline{W}$ while, clearly, $x \notin X^* \setminus \overline{W}$.

Since $X^* \setminus \overline{W}$ and $\overline{W}$ is disjoint, and $W \subseteq \overline{W}$, we have that $W \cap X^* \setminus \overline{W} = \emptyset$.

So $X^* \setminus \overline{W}$ is an open neighborhod of ∞, while W is an open neighborhood of x.

Furthermore, they are disjoint.

As x was arbitrarily chosen, we can separate any $x \in X^* \setminus \{\infty\}$ from ∞.

This together with the fact that X on its own is Hausdorff, gives Hausdorffness for X^*. ■

7 Countability- and Separation Axioms

7.1 Why do we need these axioms?

A topology can be quite large.

For instance, to know that $f : X \to Y$ is continuous, in principle we have to understand all open sets in Y. To understand what happens near x, we might have to consider all neighborhoods of x.

Countability axioms let us ask whether we can replace this potentially gigantic amount of information by a countable amount.

There are two fundamentally different notions, those being

$$\text{first countable} \;=\; \text{countably manageable locally}$$

and

$$\text{second countable} \;=\; \text{countably manageable globally.}$$

7.2 A basis at a point

Before first countability, we distinguish two concepts.

A basis for the topology is a family $\mathscr{B}$ such that every open set is a union of elements of $\mathscr{B}$, as we already know.

A **basis at a point** x, or local basis at x, is a collection $\mathscr{B}_x$ of neighborhoods of x such that for every neighborhood U of x, there exists

$$B \in \mathscr{B}$$

with

$$x \in B \subseteq U.$$

We can think of this as being arbitrarily close to x, as we often pondered in analysis.

7.3 First countability

A space X is **first countable** if every point $x \in X$ has a countable local basis.

Thus for each x,

$$\mathscr{B}_x = \{B_1, B_2, \ldots\}$$

is enough to probe every neighborhood of x.

Every metric space is firstr countable, perhaps unsurprisingly.

Indeed, at x take

$$\mathscr{B}_x = \{B_{1/n}(x) \mid n \in \mathbb{N}\}.$$

Given any neighborhood $U \ni x$, because U is metric-open there is some $\varepsilon > 0$ with

$$B_\varepsilon(x) \subseteq U.$$

Choose n sufficiently large that

$$\frac{1}{n} < \varepsilon.$$

Then

$$B_{1/n}(x) \subseteq B_\varepsilon \subseteq U.$$

So metric spaces are first countable.

7.4 Sequences

Recall that $x \in \overline{A}$ means every neighborhood of x intersects A.

Suppose there is a sequence

$$a_n \in A$$

with

$$a_n \to x.$$

Then necessarily

$$x \in \overline{A}.$$

This is because every neighborhood U of x contains all sufficiently large a_n, hence intersects A.

That direction holds in every topological space

$$a_n \in A,\ a_n \to x \Rightarrow x \in \overline{A}.$$

However, the converse fails in arbitrary topological spaces.

First countability, however, makes it hold.

7.5 The sequence lemma

If X is first countable, then

$$x \in \overline{A} \Leftrightarrow \exists a_n \in A,\ a_n \to x.$$

Let

$$B_1, B_2, \ldots$$

be a countable local basis at x.

It is convenient to make it nested,

$$C_n = \bigcap_{i \in [n]} B_i.$$

Then

$$C_1 \supseteq C_2 \supseteq C_3 \supseteq \cdots$$

and $\{C_n\}$ is still a local basis.

Since $x \in \overline{A}$, every C_n intersects A. Choose $a_n \in C_n \cap A$. Now let U be any neighborhood of x. Since $\{C_n\}$ is a local basis, some

$$C_N \subseteq U.$$

For $n \geq N$,

$$a_n \in C_n \subseteq C_N \subseteq U.$$

Therefore

$$a_n \to x.$$

7.6 Continuity can also be tested by sequences

Recall from analysis, that

$$f \text{ continuous} \Rightarrow x_n \to x \Rightarrow f(x_n) \to f(x).$$

That implication is true in arbitrary topological spaces.

But once again, the converse generally fails.

If the domain X is first countable, the conserse holds.

So all the sequential reasoning from analysis works so smoothly largely because metric spaces are first countable.

7.7 Second countability

Now we make countability global.

A space X is **second countable** if its topology has a countable basis,

$$\mathscr{B} = \{B_1, B_2, \ldots\}.$$

This is substantially stronger than first countability, and obviously implies first countability.

It should not be terribly surprising that the converse fails.

7.8 $\mathbb{R}$ is second countable

The usual topology on $\mathbb{R}$ has the basis

$$\mathscr{B} = \{(p, q) \mid p, q \in \mathbb{Q},\ p < q\}.$$

Since $\mathbb{Q} \times \mathbb{Q}$ is countable, the basis $\mathscr{B}$ is countable.

7.9 Two or more countability properties

A space X is **separable** if it has a countable dense subset

$$\exists D = \{d_1, d_2, ...\} \subseteq X$$

such that

$$\overline{D} = X.$$

Intuitively, countably many points come arbitrarily close to every point of the space.

For instance $\mathbb{Q}$ is dense in $\mathbb{R}$, so $\mathbb{R}$ is separable.

A space $X$is **Lindelöf** if every open cover has a countable subcover.

Comparatively,

$$\text{compact} \ = \ \text{every open cover has a finite subcover,}$$

while

$$\text{Lindelöf} \ = \ \text{every open cover has a countable subcover.}$$

Thus

$$\text{compact} \Rightarrow \text{Lindelöf.}$$

7.10 The central implication picture

Second countability is very strong, as it implies first countable, separable and Lindelöf.

The latter two are worth understanding.

Second countable ⇒ separable.

Let

$$\mathscr{B} = \{B_1, B_2, ...\}$$

be a countable basis.

For every nonempty B_n, choose

$$x_n \in B_n.$$

Let

$$D = \{x_n\}.$$

Then D is countable.

If $U \subseteq X$ is nonempty open, U contains some basis element B_n, and therefore

$$x_n \in U \cap D.$$

So every nonempty open set intersects D.

Hence

$$\overline{D} = X.$$

Thus X is separable.

Second countable $\Rightarrow$ Lindelöf.

Let $\mathscr{U}$ be an open cover.

For each basis element B_n that is contained in some member of $\mathscr{U}$, choose one such member

$$U_n \in \mathscr{U}.$$

There are only countably many B_n, hence only countably many selected U_n.

These cover X. If $x \in X$, choose some $U \in \mathscr{U}$ containing x. SInce $\mathscr{B}$ is a basis, there exists

$$x \in B_n \subseteq U.$$

So the selected U_n corresponding to B_n contains x.

Therefore X is Lindelöf.

7.11 In metric spaces, many of these notions collapse

For a metric space,

$$\text{second countable} \iff \text{separable} \iff \text{Lindelöf}.$$

First countability is automatic.

One useful direction is separable $\Rightarrow$ second countable.

Suppose

$$D = \{d_1, d_2, \ldots\}$$

is countable and dense.

Consider all balls

$$B_r(d),\ d \in D,\ r \in \mathbb{Q}_{>0}.$$

There are countably many and they form a basis.

7.12 Separation axioms

We now introduce a notion of how effectively a topology can distinguish different objects, with a hierarchy.

At the weakest level we can separate points and at the strongest levels we can separate a point from a closed set, as well as two disjoint closed sets.

7.13 T_1

A space X is a T_1-space if, for distinct $x, y \in X$, there exists a neighborhood of x not containing y, and vice versa.

Equivalently,

$$X \text{ is } T_1 \Leftrightarrow \{x\} \text{ is closed for every } x \in X.$$

This is because, if singletons are closed, then

$$X \setminus \{y\}$$

is an open neighborhood of x not containing y.

Conversely, if $y \neq x$ has a neighborhood not containing x, then

$$X \setminus \{y\}$$

is the union of all those neighborhoods, hence open.

Thus $\{x\}$ is closed.

7.14 Hausdorff is stronger

Hausdorff says distinct points can be separated by disjoint open neighborhoods,

$$x \neq y \Rightarrow \exists U \ni x,\ V \ni y,\ U \cap V = \varnothing.$$

Therefore Hausdorff implies T_1.

7.15 Regular spaces

In Munkres' terminology, X is regular if

1. X is T_1, and
2. whenver $x \notin C$ and $C \subseteq X$ is closed, there are disjoint open sets

$$U \ni x,\ V \supseteq C.$$

So Hausdorffness separates point from point, while regularity separates points from

closed sets.

Because singleton sets are closed in a regular space, regularity implies Hausdorffness.

7.16 Characterization of regularity

A T_1 space X is regular if and only if:

for every point x and every open neighborhood $U \ni x$, there exists and open neighborhood $V \ni x$ such that

$$\overline{V} \subseteq U.$$

We have already encountered this in locally compact Hausdorff spaces.

Why regularity gives shrinking.

Suppose $x \in U$. Then $C = X \setminus U$ is closed and does not contain x.

Regularity gives disjoint open sets

$$V \ni x,\ W \supseteq C.$$

Since $V \cap W = \emptyset$,

$$\overline{V} \cap W = \emptyset.$$

Therefore,

$$\overline{V} \subseteq X \setminus W.$$

But since

$$C = X \setminus U \subseteq W,$$

we have

$$X \setminus W \subseteq U.$$

Hence

$$\overline{V} \subseteq U.$$

7.17 Normal spaces

A T_1 space is **normal** if whenever

$$A, B \subseteq X$$

are disjoint closed sets, there exist disjoint open sets

$$U \supseteq A,\ V \supseteq B.$$

So now,

- T_1 : points individually visible,

- T_2/Hausdorff : point vs. point,
- regular: point vs. closed set,
- normal: closed set vs. closed set.

Much like regularity, we can characterize normality as "shrinking".

A T_1 space is normal if and only if, for every closed set $A \subseteq U$ where U is open, there exists an open V such that

$$A \subseteq V \subseteq \overline{V} \subseteq U.$$

7.18 Metric spaces are very well behaved (of course)

Every metric space is regular.

Suppose $x \notin C$ with C closed. Because C is closed,

$$X \setminus C$$

is open and contains x, so some ball $B_\varepsilon(x)$ avoids C.

Then something like

$$B_{\varepsilon/3}(x)$$

and the appropriate neighborhood around C can be made disjoint.

In fact, every metric space is normal.

A nice way is to use distance-to-a-set functions

$$d(x, A),\ d(x, B).$$

For disjoint closed A, B define

$$U = \{x \mid d(x, A) < d(x, B)\},\ V = \{x \mid d(x, B) < d(x, A)\}.$$

7.19 Compact Hausdorff spaces are normal

We know that compact Hausdorff spaces lets us use compactness to turn infinitely many pointwise separations into finitely many.

To separate two disjoint closed sets A, B in compact Hausdorff,

- A, B are compact since they are closed subsets of compact space,
- $\forall a \in A,\ \forall b \in B$, Hausdorffness separates a, b,
- compactness lets us make finitely many choices and combine them.

So Compact Hausdorff $\Rightarrow$ normal.

7.20 Some problems

Problem 7.1. Let X be a first-countable topological space, let $A \subseteq X$, and let $x \in \overline{A}$. Prove there exists a sequence $(a_n)_{n\in\mathbb{N}} \subseteq A$ such that

$$a_n \to x.$$

Proof. From first countability, we know that for each $x \in X$ we have a countable set

$$\mathscr{B}_x = \{B_1, B_2, \ldots\}$$

such that for every open neighborhood U of x, there is some i for which $B_i \subseteq U \neq \emptyset$.

Make the basis nested by setting

$$C_n = \bigcap_{i=1}^{n} B_i.$$

Then

$$C_1 \supseteq C_2 \supseteq C_3 \supseteq \cdots$$

and $\{C_n\}$ is still a local basis. Since $x \in \overline{A}$, every C_n intersects A. Choose $a_n \in C_n \cap A$.

Now let U be any neighborhood of x. Since $\{C_n\}$ is a local basis, some

$$C_N \subseteq U.$$

For $n \geq N$,

$$a_n \in C_n \subseteq C_N \subseteq U.$$

Therefore $a_n \to x$. ∎

Problem 7.2. Let X be first-countable and Y some topological space. Suppose $f : X \to Y$ has the property that whenever $x_n \to x$ in X, we have

$$f(x_n) \to f(x)$$

in Y.
Prove that f is continuous.

Proof. To show that f is continuous, it suffices to prove that for every point $x \in X$ and every open neighborhood V of $f(x)$ in Y, there exists an open neighborhood U of x in X such that $f(U) \subseteq V$.

We proceed by contradiction. Fix an arbitrary point $x \in X$ and assume that f is not continuous at x. Then there exists an open neighborhood V of $f(x)$ in Y such that for every open neighborhood U of x in X,

$$f(U) \nsubseteq V.$$

Equivalently, for every open neighborhood U of x, there exists a point $x_U \in U$ such that $f(x_U) \notin V$.

Since X is first-countable, the point x possesses a countable nested local basis. That is, there exists a sequence of open neighborhoods $\{U_n\}_{n=1}^{\infty}$ of x such that:

1. $U_1 \supseteq U_2 \supseteq U_3 \supseteq \dots$
2. Every open neighborhood W of x contains U_N for some $N \in \mathbb{N}$.

Using the assumption of non-continuity, for each $n \in \mathbb{N}$, we can choose a point $x_n \in U_n$ such that

$$f(x_n) \notin V.$$

We claim that the sequence $(x_n)_{n=1}^{\infty}$ converges to x in X. To verify this, let W be any open neighborhood of x. By the definition of a local basis, there exists $N \in \mathbb{N}$ such that $U_N \subseteq W$. Since the local basis is nested, for all $n \geq N$, we have

$$x_n \in U_n \subseteq U_N \subseteq W.$$

Thus, $x_n \to x$ in X.

By hypothesis, sequential continuity implies that $f(x_n) \to f(x)$ in Y. Therefore, for the open neighborhood V of $f(x)$, there must exist an integer $K \in \mathbb{N}$ such that

$$f(x_n) \in V \quad \text{for all } n \geq K.$$

However, this contradicts our choice of x_n, where $f(x_n) \notin V$ for all $n \in \mathbb{N}$.

By contradiction, f must be continuous at x. Since $x \in X$ was chosen arbitrarily, f is continuous on X. ■

Problem 7.3 (Redoing the last question)**.** Let X be **first countable**, let Y be an arbitrary topological space, and let

$$f : X \to Y.$$

Assume that for every sequence (x_n) in X,

$$x_n \to x \Rightarrow f(x_n) \to f(x).$$

Prove that f is continuous.

Proof. Suppose f is not continuous. Then there exists an open set $V \subseteq Y$ for which

$$f^{-1}(V) \subseteq X$$

is not open.

As $f^{-1}(V)$ is not open, there is at least one $x \in f^{-1}(V)$ for which x is not in its interior.

Let $x \in f^{-1}(V)$ be that point. Then

$$x \in \overline{X \setminus f^{-1}(V)}.$$

As mentioned, x is not in the interior of $f^{-1}(V)$. Thus every open neighborhood W containing $x$must intersect the complement $X \setminus f^{-1}(V)$. Thus, definitionally, x lies in the closure.

Since X is first countable, this means we can find some sequence $(x_n) \subseteq X \setminus f^{-1}(V)$ for which $x_n \to x$.

Explicitly, we can construct, using the local basis

$$\mathscr{B}_x = \{B_1, B_2, \ldots\}$$

a descending chain

$$C_n = \bigcap_{i=1}^{n} B_i \to C_1 \supseteq C_2 \supseteq C_3 \supseteq \cdots,$$

a sequence letting $x_n \in C_n \cap (X \setminus f^{-1}(V))$.

We note that then $\{C_n\}_n$ is still a local basis, and $C_n \cap (X \setminus f^{-1}(V)) \neq \emptyset$ since each C_i contains x.

We then have that $x_n \to x$ as, for any open neighborhood W of x we can find some j such that $B_j \subseteq W$. Then

$$x_n \in C_n \subseteq B_j \subseteq W$$

whenever $n \geq j$.

By hypothesis, we have that

$$f(x_n) \to f(x) \text{ in } Y.$$

Therefore, for the open neighborhood V of $f(x)$, there must exist an integer $K \in \mathbb{N}$ such that

$$f(x_n) \in V \ \forall n \geq K.$$

However, this contradicts our choice of x_n, where $f(x_n) \notin V$ for all $n \in \mathbb{N}$, since we chose x_n such that $x_n \notin f^{-1}(V)$.

So, by contradiction, f is continuous.

■

Problem 7.4. Let X be second countable. Prove that X is Lindelöf.

Proof. Suppose X is second countable.

Recall that this means that its topology has a countable basis

$$\mathscr{B} = \{B_1, B_2, \ldots\},$$

while Lindelöf means that every open cover has a countable subcover.

Suppose $\mathscr{U} = \{U_\alpha\}_\alpha$ is an arbitrary open cover of X.

Consider the basis elements B_i for which there exists U_α containing B_i. For each B_i choose one $U_{\alpha(i)}$. There are at most countably many, so this is a countable collection.

Take some $x \in X$. Since $\mathscr{U}$ covers X there is some $U_\alpha \in \mathscr{U}$ with $x \in U_\alpha$.

Since U_α is open and $\mathscr{B}$ is a basis there is some $B_i \in \mathscr{B}$ such that

$$x \in B_i \subseteq U_\alpha,$$

but then B_i was a basis element for which there was some $U_{\alpha(i)} \supseteq B_i$ so

$$x \in B_i \subseteq U_{\alpha(i)}$$

so the countable collection covers X.

■

Problem 7.5. Let X be second countable. Prove that X is separable.

Proof. A space is separable if it contains a dense countable subset, meaning a countable subset $D \subseteq X$ such that

$$\overline{D} = X.$$

Assume without loss of generality that each B_n is non-empty, discarding empty ones.

Pick a point $x_n \in B_n$ for each n (axiom of (countable) choice assumed, of course).

Let

$$D = \{x_n \mid n = 1, 2, \ldots\}.$$

D is countable. For some non-empty open set $U \subset X$ there is some basiselement $B_n \subseteq U$. Since $x_n \in B_n \subseteq U$ we have that D intersects every non-empty open set, making D dense in X. ■

Problem 7.6. Let (X, d) be a **second countable**. Prove that X is second countable.

Proof. We have a countable dense subset

$$D = \{x_1, x_2, \ldots\} \subseteq X.$$

We will construct a countable basis.

Since D is dense in X we have that D intersects every non-empty open set, so for open $U \subseteq X$ there is some $x_n \in U$.

Now, for some $x \in U$, since D is dense in X we have that some x_n lies in the open set

$$B_{r/3}(x) \subseteq U$$

for $r > 0$. Then

$$d(x, x_n) < r/3.$$

Now choose a rational q for which

$$d(x, x_n) < q < 2r/3,$$

which we can do since $\mathbb{Q}$ is dense in $\mathbb{R}$. Then

$$x \in B_q(x_n).$$

Now for any

$$y \in B_q(x_n)$$

we have

$$d(x, y) \leq d(x, x_n) + d(x_n, y) < \frac{r + 2r}{3} = r,$$

so

$$y \in B_r(x) \subseteq U,$$

meaning

$$x \in B_q(x_n) \subseteq U.$$

Thus we have a basis

$$\mathscr{B} = \{B_q(x_n) \mid n \in \mathbb{N},\ q \in \mathbb{Q}_{>0}\}$$

which is indexed over two countable indices and hence countable. Hence X is second countable. ■

Problem 7.7. Let X be a regular space.
Prove that for every $x \in X$ and every open neighborhood U of x, there exists an open set V such that

$$x \in V \subseteq \overline{V} \subseteq U.$$

Proof. Suppose X is regular, meaning it is T_1 so

$$\{x\} \text{ is closed } \forall x \in X,$$

and whenever $x \notin C$ and $C \subseteq X$ is closed, there are disjoint open sets

$$U \ni x,\ V \supseteq C.$$

We wish to prove that for every x and every open neighborhood U of x, there is an open set V such that

$$x \in V \subseteq \overline{V} \subseteq U.$$

Let U be an open neighborhood of some $x \in X$.

We have that

$$X \setminus U$$

is closed.

Thus, as $x \notin X \setminus U$, regularity gives disjoint open sets

$$V \ni x,\ W \supseteq X \setminus U.$$

Then

$$V \subseteq U$$

as if V contained some element in X not contained in U then it would intersect $X \setminus U$, contradicting disjointness.

It then remains to show that

$$\overline{V} \subseteq U.$$

Since $\overline{V}$ is the set of points y for which every open neighborhood of y intersects A.

Suppose there is some $y \in \overline{V}$ not in U. Then W would be an open neighborhood of this point, but this would imply

$$V \cap W \neq \emptyset,$$

a contradiction. Thus

$$\overline{V} \subseteq U.$$

■

Problem 7.8. Let X be a T_1-space satisfying the following property:
For every $x \in X$ and every open neighborhood U of x, there exists an open set V such that

$$x \in V \subseteq \overline{V} \subseteq U.$$

Prove that X is regular.

Proof. Suppose, for every $x \in X$ and open neighborhood U of x we can find an open set V such that

$$x \in V \subseteq \overline{V} \subseteq U.$$

We want to prove that for any x not in some closed set C we can find

$$U \ni x,\ V \supseteq C$$

where U, V are open.

Let $C \subseteq X$ be some closed set and let $x \notin C$.

Then $X \setminus C$ is an open neighborhood of x.

Thus we can find V such that

$$x \in V \subseteq \overline{V} \subseteq X \setminus C.$$

Then

$$X \setminus \overline{V}$$

is open and disjoint to V. Furthermore,

$$X \setminus \overline{V} \supseteq X \setminus U \supseteq C$$

by using standard laws of set theory:

$$\begin{aligned} &\overline{V} \subseteq U \subseteq X \setminus C \\ \Rightarrow\ &X \setminus \overline{V} \supseteq X \setminus U \supseteq C. \end{aligned}$$

■

Problem 7.9. Let X be a normal space. Prove that whenever $A \subseteq X$ is closed and $U \subseteq X$ is open with

$$A \subseteq U,$$

there exists an open set V such that

$$A \subseteq V \subseteq \overline{V} \subseteq U.$$

Proof. Recall that X normal means that X is T_1 and whenever

$$A, B \subseteq X$$

are disjoint closed sets, there exist disjoint open sets

$$U \supseteq A,\ V \supseteq B.$$

We have that $A \subseteq X$ is closed and

$$U \supseteq A$$

is open. Then

$$X \setminus U$$

is closed and disjoint to A. Thus there are disjoint open sets

$$U_A \supseteq A,\ U_{X \setminus U} \supseteq X \setminus U.$$

Since they are disjoint, we must have that

$$A \subseteq U_A \subseteq U,$$

since otherwise there would be some $x \in U_A$ for which $x \notin U$, but then

$$x \in X \setminus U \subseteq U_{X \setminus U}, \Rightarrow U_A \cap U_{X \setminus U} \neq \emptyset.$$

Now it remains to show that

$$\overline{U_A} \subseteq U.$$

Recall that $\overline{U_A}$ is the set of points y where every open neighborhood of y intersects U_A. Suppose there is some $y \in \overline{U_A}$ such that $y \notin U$.

Then every open neighborhood of y intersects U_A.

In particular, $U_{X \setminus U}$ is an open neighborhood of y, but then

$$U_{X \setminus U} \cap U_A \neq \emptyset,$$

a contradiction. Therefore, we must have

$$A \subseteq U_A \subseteq \overline{U_A} \subseteq U.$$

Problem 7.10. Let X be a T_1-space such that whenever $A \subseteq X$ is closed and $U \subseteq X$ is open with

$$A \subseteq U,$$

there exists an open set V such that

$$A \subseteq V \subseteq \overline{V} \subseteq U.$$

Prove that X is normal.

Proof. Suppose X is a T_1-space with the property above.

We wish to show that we can separate disjoint closed sets into disjoint open sets.

Suppose $A, B \subseteq X$ are disjoint closed sets.

Then $X \setminus B$ is open and

$$(X \setminus B) \supseteq A.$$

Thus there are exists some open set V for which

$$A \subseteq V \subseteq \overline{V} \subseteq X \setminus B.$$

Using standard laws of set theory,

$$X \setminus \overline{V} \supseteq B$$

and $X \setminus \overline{V}$ is open.

Then $V \supseteq A$ is disjoint to $X \setminus \overline{V}$ since $V \subseteq \overline{V}$.

Thus we have disjoint open sets

$$V \supseteq A,\ X \setminus \overline{V} \supseteq B.$$

As A, B were arbitrarily chosen closed sets, the space is normal.

■

Problem 7.11. Let X be a second-countable regular space.
Prove that X is normal.

Proof. Let X be a regular topological space with a countable basis $\mathscr{B} = \{B_1, B_2, B_3, \dots\}$. Let A and B be two disjoint closed subsets of X.

We begin by covering A with open sets whose closures do not intersect B. For each point $x \in A$, the set $X \setminus B$ is an open neighborhood of x. Since X is regular, there exists an open set U_x such that $x \in U_x$ and $\overline{U_x} \subseteq X \setminus B$.

Since $\mathscr{B}$ is a basis for X, we can find a basis element $B_n \in \mathscr{B}$ such that $x \in B_n \subseteq U_x$. Taking the closure, we have $\overline{B_n} \subseteq \overline{U_x} \subseteq X \setminus B$, which implies $\overline{B_n} \cap B = \emptyset$.

By collecting these basis elements for all $x \in A$, we obtain a countable subcollection of $\mathscr{B}$ that covers A. We index this collection as $\{U_n\}_{n\in\mathbb{N}}$. Thus, we have:

$$A \subseteq \bigcup_{n=1}^{\infty} U_n \quad \text{and} \quad \overline{U_n} \cap B = \emptyset \text{ for all } n \in \mathbb{N}.$$

By a completely symmetric argument, we can construct a countable collection of open basis elements $\{V_n\}_{n\in\mathbb{N}}$ that covers B such that:

$$B \subseteq \bigcup_{n=1}^{\infty} V_n \quad \text{and} \quad \overline{V_n} \cap A = \emptyset \text{ for all } n \in \mathbb{N}.$$

The open sets $\bigcup U_n$ and $\bigcup V_n$ are not necessarily disjoint. To eliminate overlapping points, we define two new collections of open sets, $\{U_n'\}_{n\in\mathbb{N}}$ and $\{V_n'\}_{n\in\mathbb{N}}$, by subtracting the closures of the opposing sets up to index n:

$$U_n' = U_n \setminus \bigcup_{i=1}^{n} \overline{V_i} \quad \text{and} \quad V_n' = V_n \setminus \bigcup_{i=1}^{n} \overline{U_i}$$

Since each $\overline{V_i}$ and $\overline{U_i}$ is closed, their finite unions are closed. Therefore, U_n' and V_n' are open sets.

Now, define U and V as the unions of these modified sets:

$$U = \bigcup_{n=1}^{\infty} U_n' \quad \text{and} \quad V = \bigcup_{n=1}^{\infty} V_n'$$

Being unions of open sets, both U and V are open. We now verify that U and V satisfy the conditions for normality:

1. **Containment ($A \subseteq U$):** Let $x \in A$. Since $\{U_n\}$ covers A, there is some integer n such that $x \in U_n$. Furthermore, since $\overline{V_i} \cap A = \emptyset$ for all i, the point x cannot belong to $\bigcup_{i=1}^{n} \overline{V_i}$. Thus, $x \in U_n \setminus \bigcup_{i=1}^{n} \overline{V_i} = U_n' \subseteq U$. Hence, $A \subseteq U$.
2. **Containment ($B \subseteq V$):** By a symmetric argument, if $y \in B$, then $y \in V_n$ for some n. Since $\overline{U_i} \cap B = \emptyset$ for all i, $y \notin \bigcup_{i=1}^{n} \overline{U_i}$. Thus, $y \in V_n' \subseteq V$. Hence, $B \subseteq V$.
3. **Disjointness ($U \cap V = \emptyset$):** Suppose for contradiction that there exists a point $z \in U \cap V$. Then $z \in U_n'$ for some $n \in \mathbb{N}$ and $z \in V_m'$ for some $m \in \mathbb{N}$. We analyze the two possible relations between indices n and m:
   - If $n \leq m$: By definition, $V_m' = V_m \setminus \bigcup_{i=1}^{m} \overline{U_i}$. Since $n \leq m$, $\overline{U_n}$ is included in this subtracted union, meaning $V_m' \cap \overline{U_n} = \emptyset$. Since $z \in U_n' \subseteq U_n \subseteq \overline{U_n}$, z cannot be in V_m', which is a contradiction.
   - If $m < n$: By definition, $U_n' = U_n \setminus \bigcup_{i=1}^{n} \overline{V_i}$. Since $m < n$, $\overline{V_m}$ is included in this subtracted union, meaning $U_n' \cap \overline{V_m} = \emptyset$. Since $z \in V_m' \subseteq V_m \subseteq \overline{V_m}$, z cannot be in U_n', which is a contradiction.

   In either case, a contradiction arises. Thus, $U \cap V = \emptyset$.

Since we have found disjoint open sets U and V such that $A \subseteq U$ and $B \subseteq V$, the space X is normal. ■

> **Problem 7.12.** Let X be a **regular Lindelöf space**.
> Prove that X is normal.

Proof. X is regular Lindelöf, so regularity gives that for $x \notin D$ for some closed $D \subseteq X$ we can find disjoint open

$$U \ni x,\ V \supseteq D,$$

while Lindelöf says that every conver can be reduced to a countable subcover.

Let $A, B \subseteq X$ be closed sets. Then $X \setminus B$ is an open neighborhood of x for which, by regularity, we can find open V_x such that

$$x \in V_x \subseteq \overline{V_x} \subseteq X \setminus B.$$

Then

$$\bigcup_{x \in A} V_x$$

is a cover of A with

$$\left(\bigcup_{x\in A} V_x\right) \cap B = \emptyset.$$

Since X is Lindelöf, we can reduce this to a countable subcover

$$\bigcup_{x(n)\in A} V_x$$

still disjoint to B.

Now, for each $\in B$, regularity gives an open W_y with

$$y \in W_y \subseteq \overline{W_y} \subseteq X \setminus A.$$

Reduce this to a countable cover of B,

$$\bigcup_{y(m)\in B} W_y$$

disjoint to A.

Now these need not be disjoint.

So set

$$V_n' = V_{x(n)} \setminus \bigcup_{i=1}^{n} \overline{W_{y(i)}} \text{ and } W_n' = W_{y(n)} \setminus \bigcup_{i=1}^{n} \overline{V_{x(i)}}.$$

Recalling then that

$$V_n' = V_{x(n)} \cap \bigcap_{i=1}^{n} X \setminus \overline{W_{y(i)}} \text{ and } W_n' = W_{y(n)} \cap \bigcap_{i=1}^{n} X \setminus \overline{V_{x(i)}}$$

which is open.

Then, for each of these,

$$A \subseteq \bigcup_{n=1}^{\infty} V_n', \text{ and } B \subseteq \bigcup_{n=1}^{\infty} W_n'$$

since we only removed elements which were disjoint to A and B anyways. Furthermore, these sets are open since arbitary unions of open sets are open.

These sets are disjoint by design since if we suppose $z \in V_n' \cap W_m'$, then if $n \leq m$ then W_m' was defined by removing $\overline{V_n}$, but

$$z \in V_n' \subseteq V_n \subseteq \overline{V_n},$$

a contradiction. Applying the symmetric argument for $m < n$ we get a similar contradiciton so the collections are disjoint.

Problem 7.13. Let X be un uncountable set equipped with the discrete topology. Determine which of the following properties X has, and prove every claim:

$$T_1,\ \text{Hausdorff},\ \text{regular},\ \text{normal},$$
$$\text{first countable},\ \text{second countable},\ \text{separable},\ \text{Lindelöf}.$$

Solution and proof.

We will have a lot of opportunities here to assert that the space does or does not have some property using implications from properties we prove along the way, but for simplicity we will prove or disprove all claims from their definition.

T_1.

The discrete topology makes X T_1 since for distinct $x, y \in X$ we have that

$$X \setminus \{y\}$$

is an open set, containing x and not containing y.

Hausdorff.

The space is indeed Hausdorff.

Take distinct $x, y \in X$. Then $\{x\},\ \{y\}$ are open neighborhoods of x and y with

$$\{x\} \cap \{y\} = \emptyset.$$

Regular.

Take $C \subseteq X$ to be some closed set not containing $x \in X$.

We want to know whether we can always find some open sets $U \ni x$ and $V \supseteq C$ such that

$$U \cap V = \emptyset.$$

We can very obviously do this since $x \in \{x\}$ and $X \setminus \{x\}$ are open sets satisfying the condition.

Normal.

Once again this is quite trivial.

Suppose $A, B \subseteq X$ are disjoint closed sets.

Then

$$A \supseteq A,\ B \supseteq B$$

are disjoint open sets containing A and B, respectively.

First countable.

The space is first countable.

Take

$$\mathscr{B}_x = \{\{x\}\}$$

for all $x \in X$.

Then, for fixed $x \in X$, any neighborhood of of x intersects $\{x\}$ by definition.

Second countable.

The space is not second countable.

Some basis $\mathscr{B}$ must be able to recreate every open set, but

$$\{x\}$$

is open, so $\mathscr{B} \ni \{x\}$ for every x, of which there are uncountably many.

Separable.

A space is separable if it contains a dense countable subset, meaning we need some countable $D \subseteq X$ for which

$$\overline{D} = X.$$

Since X has the discrete topology, the closure of any set is the set itself.

For a closed set $Y \subseteq X$, Y is itself open.

And for an open set $Y \subseteq X$, Y is closed since $X \setminus Y$ is open.

Thus the only subset $D \subseteq X$ for which $\overline{D} = X$ is X itself, which is uncountable.

Lindelöf.

The space is not Lindelöf.

We have that

$$\bigcup_{x \in X} \{x\} = X,$$

but if we remove any $\{x\}$ it won't cover the space anymore so this cannot be reduced to a countable subcover.

■

Problem 7.14. Let X be a compact Hausdorff space.
Prove that X is normal.

Proof. X is Hausdorff so for distinct $x, y \in X$ we can find open

$$U_x \ni x,\ U_y \ni y,$$

disjoint.

The space is compact so every cover has a finite subcover.

We want to prove that the space is normal, meaning for disjoint closed $A, B \subseteq X$, we can find disjoint open sets

$$U_A \supseteq A,\ U_B \supseteq B.$$

Now, let $A, B \subseteq X$ be closed sets.

For $x \in A$ and $y \in B$ these are distinct since A, B are disjoint.

So we can find disjoint open sets

$$U_x \ni x,\ U_y \ni y.$$

In particular, for $x \in A$ and every $y \in B$ we can find disjoint open sets

$$U_x \ni x,\ U_{x,y} \ni y.$$

The collection $\{U_x\}_{x \in A}$ forms an open cover of A. Reduce this to a finite cover

$$U = \bigcup_{i=1}^{n} U_{x_i} \supseteq A.$$

Corresponding to each U_{x_i} we have the open set

$$V_{x_i} = \bigcup_{y \in B} U_{x,y},\ x \in U_{x_i}.$$

Defibne the open set

$$V = \bigcap_{i=1} V_{x_i}.$$

We have $B \subseteq V$ and V and U are disjoint since

$$U \cap V = \left(\bigcup_i U_{x_i}\right) \cap \left(\bigcap_j V_{x_j}\right) \subseteq \bigcup_{i=1}^{n} (U_{x_i} \cap V_{x_i}) = \emptyset.$$

■

Problem 7.15. Let X be a **second-countable regular space**, and let

$$Y \subseteq X$$

be an arbitrary subspace.
Prove that Y is normal.

Proof. X is second countable, so it has a countable basis

$$\mathscr{B} = \{B_1, B_2, \ldots\},$$

and it is regular so for $C \subseteq X$ closed and $x \notin C$, we have that there exist disjoint open sets U_x, U_C for which

$$U_x \ni x, \text{ and } U_C \supseteq C.$$

We wish to prove that for disjoint closed subsets of Y, we can divide them into disjoint open sets.

We first note that Y is second countable, with basis

$$\mathscr{B}_Y = \{B_1 \cap Y, B_2 \cap Y, ...\},$$

and Y is regular.

For regularity we note that X is T_1 and thus Y is T_1.

Let $F \subseteq Y$ be closed with $y \in Y \setminus F$. Then $F = Y \cap C$ for some closed set $C \subseteq X$ so if $y \notin F$ then $y \notin C$. Since X is regular there are disjoint open sets U_y, U_C containing y and C, respectively. Then $U_y \cap Y$ and $U_C \cap Y$ are disjoint open neighborhoods in Y with y and F, respectively.

Y is also Lindelöf, since second countable implies Lindelöf.

Lastly, second countable + Lindelöf implies normal, as we have proven previously. ■

8 Some recap before doing last part of point-set Munkres and proceeding to algebraic topology

Problem 8.1. Let $q : X \to Y$ be a surjective map, where X is a topological space. Suppose Y is equipped with the quotient topology induced by q.
Let Z be another topological space and let

$$g : Y \to Z$$

be a function.
Prove that g is continuous if and only if $g \circ q : X \to Z$ is continuous.

Proof. The quotient topology is the finest topology making q continuous. I.e.

$$A \subseteq Y \text{ open} \Leftrightarrow f^{-1}(A) \text{ open}.$$

Suppose g is continuous. Then, since q is continuous, $g \circ q$ is continuous as composition preserves continuity.

Now suppose $g \circ q : X \to Y$ is continuous.

Then for some open set $B \subseteq X$, we have that

$$(g \circ q)^{-1}(B) \subseteq X$$

is open.

Now,

$$(g \circ q)^{-1} = q^{-1} \circ g^{-1}.$$

So

$$(g \circ q)^{-1}(B) = q^{-1}(g^{-1}(B)).$$

By virtue of Y having the quotient topology,

$$q^{-1}(g^{-1})(B)$$

open, implies that

$$q(q^{-1}(g^{-1}(B))) = g^{-1}(B)$$

is open. As $B \subseteq Z$ was arbitrary,

$$g^{-1}(B)$$

is open for any open $B \subseteq Z$, making g continuous. ■

Problem 8.2. Let $q : X \to Y$ be a quotient map, and let $A \subseteq X$ be saturated. Assume A is open in X.
Prove that

$$q(A)$$

is open in Y.

Proof. We have

$$A = q^{-1}(q(A)).$$

By the property of the quotient map, $B \subseteq Y$ is open if and only if $q^{-1}(B) \subseteq X$ is open.

Set $B := q(A)$. We have

$$q^{-1}(q(A)) = A,$$

which is open in X, hence

$$q(A)$$

is open in Y. ■

Problem 8.3. Let $q : X \to Y$ be a quotient map.
Suppose $C \subseteq Y$ is closed. Prove that

$$q^{-1}(C)$$

is closed in X.

Then determine if the converse is always true,

$$q^{-1}(C) \text{ closed in } X \overset{?}{\Rightarrow} C \text{ closed in } Y.$$

Proof. Suppose $C \subseteq Y$ is closed. Then $C = Y \setminus A$ for some open set $A \subseteq Y$.

Then

$$q^{-1}(A) \subseteq X$$

is open, but then

$$q^{-1}(C) = q^{-1}(Y \setminus A) = X \setminus q^{-1}(A)$$

which is closed in X.

Suppose $q^{-1}(C)$ is some closed set. Then

$$X \setminus q^{-1}(C)$$

is open, but

$$X \setminus q^{-1}(C) = q^{-1}(Y \setminus C).$$

Therefore $q^{-1}(Y \setminus C)$ is open in X. Since q is a quotient map,

$$Y \setminus C$$

is open, making C closed.

■

Problem 8.4. Let $q : X \to Y$ be a quotient map and let $A \subseteq X$.
Prove that

$$q^{-1}(q(A))$$

is the smallest saturated subset of X containing A.

Proof. We first note that $q^{-1}(q(A))$ is a saturated subset of X containing A.

For containment, let $x \in A$. Then

$$q(x) \in q(A),$$

but

$$q^{-1}(q(A)) = \{x \in A \mid x \in q(A)\}$$

so

$$x \in q^{-1}(q(A)).$$

Thus

$$A \subseteq q^{-1}(q(A)).$$

For saturation, we prove the backwards direction, since the forward one is trivial.

Set $C := f^{-1}(f(A))$.

Let $x \in C$. Then

$$f(c) \in f(C).$$

Since $C = f^{-1}(f(A))$ we have

$$f(x) \in f(f^{-1}(f(A))).$$

Notice that for any set A, the image of its preimage satisfies $f(f^{-1}(f(A))) = f(A)$.

Therefore $f(x) \in f(A)$. By the definition of the preimage, since $f(x) \in f(A)$, we hav $x \in f^{-1}(f(A)) = C$.

So C is saturated.

Now suppose some other set is saturated and containing A, meaning we have some $B \subseteq X$ for which $A \subseteq B$ with

$$B = f^{-1}(f(B)).$$

Let $x \in C = f^{-1}(f(A))$.

Then, as

$$A \subseteq B,$$

we have that

$$f(x) \in f(B).$$

Since B is saturated, we then have that

$$x \in f^{-1}(f(x)) \subseteq f^{-1}(f(B)) = B.$$

Thus

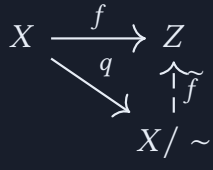
$$C \subseteq B,$$

meaning C is the smallest saturated set containing A. ■

Problem 8.5. Let $q : X \to X/\sim$ be the quotient map associated to an equivalence relation.
Suppose $f : X \to Z$ is continuous and constant on equivalence classes.
Prove there exists a unique continuous map $\widetilde{f} : X/\sim \to Z$ such that

$$f = \widetilde{f} \circ q.$$

Proof. We wish to prove, that there exists a unique map $\widetilde{f} : X/\sim \to Z$ making



commute.

Suppose $f : X \to Z$ is constant on equivalence classes. For

$$f(x) = z$$

the map $\widetilde{f} : X/\sim \to f$, by

$$\widetilde{f}([x]) = z$$

is well-defined, since f is constant on equivalence classes.

Then,

$$f = \widetilde{f} \circ q,$$

since

$$f(x) = z$$

and

$$(\widetilde{f} \circ q)(x) = \widetilde{f}([x]) = z.$$

The map is continuous by the universal property, but without that we can observe that since if $A \subseteq Z$ is open, then

$$f^{-1}(A) = (\widetilde{f} \circ q)^{-1} = q^{-1}(\widetilde{f}^{-1}(A))$$

is open. Since q has the property that $q^{-1}(B)$ is open if and only if B is open, it must be the case that $\widetilde{f}^{-1}(A)$ is open in $X/\sim$.

What remains, is to prove that this map is unique.

Suppose there exists some other map $\ell : X/\sim \to Z$ making the triangle above commute.

Then

$$\widetilde{f} \circ q = f = \ell \circ q.$$

Then, for every $x \in X$ we have

$$\widetilde{f}(q(x)) = f(x) = \ell(q(x)).$$

Therefore,

$$\widetilde{f}(q(x)) = \ell(q(x)), \ \forall x \in X.$$

Since the map is surjective we can write $y = q(x)$ for every $y \in X/\sim$ for some $x \in X$.

Eveluating the functions at y, give

$$\widetilde{f}(y) = \widetilde{f}(q(x)) = \ell(q(x)) = \ell(x).$$

Thus $\widetilde{f} = \ell$, so the map is unique.

■

Problem 8.6 (Problem for Håkon)**.** Let X be a compact space, and Y a Hausdorff space.
Let $f : X \to Y$ be a continuous bijective function.
Prove that f is a homeomorphism, meaning

$$X \cong Y.$$

Proof. Since X is compact, any cover

$$\bigcup_{\alpha} U_\alpha = X.$$

We will prove that for a closed set $C \subseteq X$, the image

$$f(C) \subseteq Y$$

is closed. Then we will conclude that f^{-1} (which exists, since f is bijective) is continuous.

Take $C \subseteq X$ to be closed. Then C is compact, since closed subspaces of compact spaces are compact.

Then

$$f(C) \subseteq Y$$

is compact, since continuous functions preserve compactness.

Thus $f(C)$ is closed, since compact subspaces of Hausdorff spaces are closed.

As f is a closed function, f^{-1} is continuous.

Thus f is a homeomorphism. ∎

Problem 8.7. Define an equivalence relation on $[0, 1]$ by identifying $0 \sim 1$, with all other equivalence classes trivial.
Let $X = [0, 1]/\sim$. Prove that

$$X \cong S^1.$$

Proof. Set theoretically, the 1-sphere, S^1, is defined by

$$S^1 = \{(x, y) \in \mathbb{R}^2 \mid x^2 + y^2 = 1\}.$$

Let $f : [0, 1] \to S^1$, by

$$f(x) = (\cos(2\pi x), \sin(2\pi x)).$$

Then f is constant on equivalence classes since

$$f(0) = 0 \equiv_{2\pi} 2\pi = f(1).$$

This map is obviously continuous.

Thus this induces a unique continuous map $\widetilde{f} : X \to S^1$.

We note that X is compact and S^1 is Hausdorff.

Since $\widetilde{f}^{-1}$ by

$$\widetilde{f}^{-1}(2\pi) = \widetilde{f}^{-1}(0) = [0] = [1]$$

with the fibers of all other elements singleton classes being

$$\widetilde{f}^{-1}(\cos\theta, \sin\theta) = \frac{\cos\theta + 1}{2}$$

writing $(x, y) = (\cos\theta, \sin\theta)$.

We have that the $\widetilde{f}$ is continuous bijective. Since X is compact, and S^1 is Hausdorff, the function $\widetilde{f}$ is a homeomorphism. ■

Problem 8.8. Let X be a locally connected space, and let C be a connected component of X.
Prove that C is open in X.

Proof. We suppose C is a maximal connected subset $C \subseteq X$.

Since the space is locally connected, we can, for every neighborhood U of $x \in X$, find a connected open neighborhood V such that

$$x \in V \subseteq U.$$

Since X is open we can find, for each $x \in C$,

$$U_x \subseteq X$$

open, and connected.

Now $x \in C \cap U_x$ so

$$U_x \cup C$$

is connected, but C is a maximal connected set so

$$U_x \subset C.$$

Since this holds for all $x \in C$, we have

$$C = \bigcup_{x \in C} U_x$$

which is open.

■

Problem 8.9. Let X be **locally path connected**.
Prove that every connected component of X is path connected.
Equivalently, prove that in a locally path connected space, connected components $=$ path components.

Proof. Suppose $C \subseteq X$ is some connected component.

Fix $x \in C$.

Since X is path connected there exists a path-connected open neighborhood U of x such that

$$x \in U \subseteq X.$$

Since U is path connected and contains x we must have that every point in U can be joined by a path to x. Since $x \in C$ it follows that

$$U \subseteq C$$

since U must lie entirely in the same component as x.

■

9 §32, §33, and §34

Under Munkres' convention, X is normal if it is T_1 and any two disjoint closed sets A, B have disjoint open neighborhoods.

The point of §32 is mostly to establish large classes of normal spaces. The important theorems we cover here are

$$\text{regular + countable basis} \implies \text{normal},$$

$$\text{metrizable} \implies \text{normal},$$

$$\text{compact Hausdorff} \implies \text{normal}.$$

Munkres also proves normality for well-ordered sets in the order topology.

These have been proven in problems earlier so yeah, that's why I largely skip it here.

9.1 §33 - Urysohn's lemma

Theorem 9.1. *If X is normal and $A, B \subseteq X$ are disjoint closed sets, then there exists a continuous*

$$f : X \to [0, 1]$$

> *such that*
>
> $$f(A) = \{0\}, \text{ and } f(B) = \{1\}.$$

Geometrically, we may interpret normality as A and B being able to be pulled apart by open sets.

Urysohn is stronger, saying there is an entire continuous numerical coordinate that distinguishes them by

$$A \mapsto 0,\ B \mapsto 1.$$

We should think of $f(x)$ as measuring, topologically rather than metrically, "how far through the transition from A to B" x lies.

This is quite nice, because normality gives

$$A \subseteq U \subseteq \overline{U} \subseteq X \setminus B,$$

while a continuous function into $[0, 1]$ requires infinitely many intermediate levels.

So Urysohn's idea is to repeatedly insert more open sets between existing ones.

Start with something corresponding to $0 < 1$.

Normality lets us choose open U_0, U_1 so that

$$A \subseteq U_0,\ \overline{U_0} \subseteq U_1 = X \setminus B.$$

Now insert a set at $1/2$ so that

$$\overline{U_0} \subseteq U_{1/2} \subseteq \overline{U_{1/2}} \subseteq U_1.$$

Then insert $1/4$ and $3/4$ to get

$$U\overline{U_0} \subseteq U_{1/4} \subseteq \overline{U_{1/4}} \subseteq U_{1/2}$$

and

$$\overline{U_{1/2}} \subseteq U_{3/4} \subseteq \overline{U_{3/4}} \subseteq U_1.$$

Continue recursively over the dyadic rationals

$$0, \frac{1}{2}, \frac{1}{1}, \frac{1}{4}, \frac{3}{4}, \frac{1}{8}, \dots$$

until you have an open set U_r satisftying the crucial property

$$r < s \Rightarrow \overline{U_r} \subseteq U_s.$$

That closure is the important part since mere nesting $U_r \subseteq U_s$ is not strong enough to force continuity later.

9.1.1 Construction of the function

Once all U_r exist, define roughly

$$f(x) = \inf\{r \mid x \in U_r\},$$

with the value 1 for points belonging to none of the relevant U_r's.

The nesting means that points deep toward A enter U_r early, so $f(x)$ is small, and points toward B enter late or never so $f(x)$ is large.

In particular,

$$f(a) = 0 \ (a \in A)$$

and

$$f(b) = 1 \ (b \in B).$$

The clever bit is proving continuity from

$$\overline{U_r} \subseteq U_s, \ (r < s).$$

For instance if

$$f(x) < a$$

you can choose a dydadic rational $r < a$ with $x \in U_r$ and the entire open neighborhood U_r consists of points whose f values stay below a.

Similarly, if $f(x) > a$, you choose an appropriate rational level below $f(x)$ and use the complement of a closure

$$X \setminus \overline{U_r}$$

as an open neighborhood on which f stays above a.

9.1.2 Reformulating normality

Urysohn gives

$$X \text{ normal} \Rightarrow \text{ disjoint closed sets can be separated by } f : X \to [0,1].$$

The converse is nearly immediate. Suppose every pair of closed sets A, B admit f with $f(A) = 0$ and $f(B) = 1$. Then, e.g.

$$U = f^{-1}([0, 1/3)),$$

and

$$V = f^{-1}((2/3, 1])$$

work.

10 Urysohn metrization

Now we get to the punchline of these sections, that being that,

> **Theorem 10.1** (Urysohn Metrization Theorem). *Every regular space with countable basis is metrizable.*

This looks quite surprising since a metric $d : X \times X \to \mathbb{R}$ seems like extra structure, while Urysohn's theorem simply asks for two purely topological conditions.

WE already knew that

$$\text{metric} \implies \text{regular},$$

but not every metric space is second countable.

10.1 How do we get a metric?

The procedure seems a bit nonconstructive, as we do not try to come up with a $d(x, y)$ first.

Instead, we make lots of continuous functions

$$f_n : X \to [0, 1]$$

which collectively distinguish points and then send

$$x \mapsto (f_1(x), f_2(x), \ldots)$$

giving a map

$$F : X \to [0, 1]^{\mathbb{N}},$$

but we already know that the countable product of $[0, 1]$ with the product topology is metrizable.

Therefore, if F is an embedding,

$$X \cong F(X) \subseteq [0, 1]^{\omega}$$

and as a subspace of a metric space, X is metrizable.

We get the family $f_1, f_2, \ldots$ as follows.

Let

$$\mathscr{B} = \{B_1, B_2, \ldots\}$$

be a countable basis and look at pairs (B_m, B_n) such that

$$\overline{B_m} \subseteq B_n.$$

There are only countably many pairs since $\mathbb{N}^2$ is countable. For each appropriate pair,

consider the closed disjoint sets

$$\overline{B_m}$$

and

$$X \setminus B_n.$$

Since the space is regular and second countable, it is normal. Then Urysohn gives a continuous function

$$f_{m,n} : X \to [0,1]$$

such that, say

$$f_{m,n} = 1 \text{ on } \overline{B_m},$$

and

$$f_{m,n} = 0 \text{ on } X \setminus B_n.$$

10.2 Some problems

Problem 10.1. Let (X,d) be a metric space and let $A, B \subseteq X$ be disjoint closed sets. Prove that X is normal.

Proof. We first note that X is T_1 Hausdorff since all metric spaces are T_1 Hausdorff.

We will use the new Urysohn approach to show normality.

We will construct a function $f : X \to [0,1]$ continuous for which

$$f(A) = 0$$

and

$$f(B) = 1.$$

Set

$$f(x) = \frac{d(x,A)}{d(x,A) + d(x,B)}$$

with

$$d(x,A) := \inf_{a \in A} d(x,a).$$

Then, for any $x \in A$,

$$f(x) = \frac{0}{0 + d(x,B)} = 0$$

and, for any $x \in B$,

$$f(x) = \frac{d(x,A)}{d(x,A) + 0} = 1.$$

This function is continuous since the distance-to-set function is continuous and $f + g$ as well as h/j are continuous for continuous f, g, h, j.

We have not proven well-definedness, but it should be clear that $f(x) \in \{0,1\}$ only if

$x \in A$ or $x \in B$ and otherwise, for $x \notin A \cup B$ we have $d(x, A) > 0$ and $d(x, A) + d(x, B) > d(x, A)$ with both sides nonzero so

$$f(x) < 1.$$

Thus the space is normal. ∎

11 §51, Moving towards Algebraic Topology

Finally, we begin some algebraic topology.

11.1 Homotopy of paths

Let $f, g : X \to Y$ be continuous.

We say f and g are **homotopic**, usually writing $f \simeq g$ if if there exists a continuous map

$$F : X \times I \to Y, \ I = [0, 1],$$

such that

$$F(x, 0) = f(x), \ F(x, 1) = g(x), \ \forall x \in X.$$

The parameter $t \in I$ may be thought of as a time parameter.

For each fixed t,

$$F_t(x) = F(x, t)$$

is an intermediate map $X \to Y$, so we can picture

$$f = F_0 \rightsquigarrow F_{0.1} \rightsquigarrow F_{0.2} \rightsquigarrow \cdots \rightsquigarrow F_1 = g.$$

The requirement that $F : X \times I \to Y$ is continuous is what formalizes "continuous deformation".

11.2 Staright-line homotopies

The easiest example is when $Y \subseteq \mathbb{R}^n$ is convex.

Given

$$f, g : X \to Y,$$

define

$$F(x, t) = (1 - t)f(x) + tg(x).$$

At $t = 0$,

$$F(x, 0) = f(x),$$

and at $t = 1$,

$$F(x, 1) = g(x).$$

Convexity just gurantees

$$(1-t)f(x)+tg(x)\in Y.$$

So any continuous functions into a convex space are homotopic by a straight-line homotopy.

So the shape of the codomain is important since something like $\mathbb{R}^2\setminus\{0\}$ causes issues. A straight-line deformation through the origin would not be well-defined.

11.3 Paths are maps

A path from x_0 to x_1 is simply a continuous map

$$f:I\to X$$

such that

$$f(0)=x_0,\ f(1)=x_1.$$

So homotopy lets us take this a level up and ask whether two paths are essentially the same by asking about path homotopy.

11.4 Path homotopy

Suppose $f,g:I\to X$ are both paths. They are **path homotopic**, written

$$f\simeq_p g,$$

if there exists continuous

$$F:I\times I\to X$$

such that

$$F(s,0)=f(x),\ f(s,1)=g(s),$$

and crucially,

$$F(0,t)=x_0,\ F(1,t)=x_1$$

for every $t\in I$.

This distinguishes path homotopy from arbitrary homotopy as we can picture the square $I\times I$ with the bottom edge mapping to f,

$$F(s,0)=f(s)$$

and the top edge mapping to g,

$$F(s,1)=g(s)$$

with the left and right edges remaining frozen,

$$F(0,t)=x_0,\ F(1,t)=x_1.$$

11.5 Fixing endpoints

Suppose we did not fix the endpoints.

Then two paths could be called equivalent while their endpoints wander all over X which would be disasterous for concatenating paths later, because the endpoint information is what tells us whether one path can follow another.

Path homotopy preserves source x_0 and target x_1, indicative of category/groupid-esque ideas.

11.6 Path homotopy is an equivalence relation

Munkres proves both ordinary and path homotopy to be equivalence relations.

Reflexivity.

For a path f, define

$$F(s,t) = f(s).$$

Nothing changes with t so

$$f \simeq_p f.$$

Symmetry.

If F deforms f into g, define

$$G(s,t) = F(s, 1-t).$$

This runs the homotopy backwards so

$$G(s,t) = F(s, 1-t).$$

Thus

$$g \simeq_p f.$$

Reflexivity.

Suppose

$$f \simeq_p g$$

via F, and

$$g \simeq_p h$$

via G.

Run F during the first half and G during the second to get

$$H(s,t) = \begin{cases} F(s,2t), & 0 \le t \le \frac{1}{2}, \\ G(s,2t-1), & \frac{1}{2} \le t \le 1. \end{cases}$$

At $t = 1/2$, both pieces give $g(s)$.

Applying the **pasting lemma** gives continuity.

11.7 Equivalence classes of paths

Because $\simeq_p$ is an equivalence relation, we can form classes

$$[f].$$

Instead of caring about a specific parameterized path f we care about its homotopy class, hinting at the first big algebraic topology esque idea, replacing geometric objects by equivalence classes.

11.8 Trivial case with convex spaces

Suppose $X \subseteq \mathbb{R}^n$ is convex, and f, g are two paths from x_0 to x_1.

Then

$$F(s,t) = (1-t)f(s) + tg(s)$$

is a path homotopy.

So in a convex space, any two paths with the same endpoints are homotopic.

Contrast that with the puncutred plane $\mathbb{R}^n \setminus \{0\}$.

Intuitively, a path going above the hole cannot be continuously deformed to a path going below due to intermediate value theorem type behavior. There would have to be a continuous path going through the origin which would not be well defined.

This is getting at homotopys ability to detect holes in the space.

11.9 Multiplying paths

Suppose $f : I \to X$ goes from x_0 to x_1 and $g : I \to X$ goes from x_1 to x_2.

We want to travel along f then g.

Define

$$f * g : I \to X$$

by

$$(f*g)(s) = \begin{cases} f(2s), & 0 \le s \le \frac{1}{2}, \\ g(2s-1), & \frac{1}{2} \le s \le 1. \end{cases}$$

This is called the path product.

At the joining point we have

$$f(1) = x_1 = g(0)$$

so this is well defined.

11.10 Why we multiply classes instead of paths

We would like

$$[f] * [g] = [f * g]$$

but we have to check well-definedness.

Suppose

$$f \simeq_p f'$$

and

$$g \simeq_p g'.$$

We need

$$f * g \simeq_p f' * g'.$$

If F is a homotopy from f to f' and G from g to g', define

$$H(s,t) = \begin{cases} F(2s,t), & 0 \le s \le \frac{1}{2} \\ G(2s-1,t) & \frac{1}{2} \le s \le 1. \end{cases}$$

Then H is the required homotopy. SO

$$[f] * [g]$$

depends only on the classes, not the chosen representatives.

11.11 Subtelty with associativity

We might expect

$$(f * g) * h = f * (g * h),$$

but as actual functions $I \to X$, this is generally false.

We roughly get

$$\underbrace{(f * g)}_{\text{first half}} * \underbrace{h}_{\text{second half}}$$

so f is the first quarter while g is the second quarter, while

$$\underbrace{f}_{\text{first half}} * \underbrace{(g * h)}_{\text{second half}}$$

so h, in this context, is the last quarter.

So they trace the same route, geometrically, but with different parameterizations.

However, they are path homotopic, meaning

$$([f] * [g]) * [h] = [f] * ([g] * [h]).$$

11.12 Indetify paths

For $x \in X$, define the constant path

$$e_x(s) = x.$$

If f goes from x_0 to x_1, then

$$e_{x_0} * f$$

can be interpreted as standing still at x_0 for half the alloted time before running along f which is not literally the same parameterized path as f, but it is path homotopic to f.

Similarly,

$$f * e_{x_1} \simeq_p f.$$

Thus on equivalence classes,

$$[e_{x_0}] * [f] \simeq [f] \simeq [f] * [e_{x_1}]$$

which is starting to smell very much **"like a group"**.

11.13 Reverse paths

Given $f : I \to X$, define the reverse path

$$\overline{f}(s) = f(1 - s).$$

If f goes $x_0 \to x_1$, then $\overline{f}$ goes $x_1 \to x_0$. Going out and back should be the same as doing nothing so

$$f * \overline{f} \simeq_p e_{x_0}$$

with

$$\overline{f} * f \simeq_p e_{x_1}.$$

Therefore

$$[f] * [\overline{f}] = [e_{x_0}]$$

and

$$[\overline{f}] * [f] = [e_{x_1}].$$

11.14 Not quite a group

We now have

- objects: points of X,
- arrows $x \to y$: path-homotopy classes of paths from x to y,
- composition:

$$[f] * [g],$$

- identifies:
$$[e_x],$$
- with every arrow having an inverse $[\overline{f}]$,

but composition is only defined when endpoints match.

This structure is not quite a group, but rather a so-called **groupoid**, giving a very natural viewpoint,
$$\Pi_1(X) = \text{fundamental groupoid of } X.$$

11.15 The group in §52

Choose a basepoint $x_0 \in X$. Now consider only paths satisfying $f(0) = f(1) = x_0$.

These are **loops based at** x_0.

If f, g are both loops based at x_0 then
$$f(1) = x_0 g(0)$$
so $f * g$ is always defined.

Thus on path-homotopy classes of based loops we have

- everywhere based binary operation,
- associativity,
- identity $[e_{x_0}]$,
- inverse $[\overline{f}]$,

i.e. we get a group!

In §52 we will discuss **the fundamental group**
$$\pi_1(X, x_0).$$

12 §52

Yippe, finally the fundamental group. Recall from §51 that if f, g are paths, then $f * g$ is defined only if $f(1) = g(0)$. That gave a groupoid rather than a group.

Now, fix one point $x_0 \in X$ and consider only paths that begin and end there.

Such a path
$$f : I \to X,\ f(0) = f(1) = x_0$$
is a **loop** based at x_0.

We then quotient the loops by path homotopy relative to the endpoints.

The resulting set is

$$\pi_1(X, x_0) = \{\text{path homotopy classes of loops based at } x_0\}.$$

The operation is $[f][g] = [f * g]$. From §51 this is well defined and satisfies

$$([f][g])[h] = [f]([g][h]),$$

$$[e_{x_0}][f] = f = [f][e_{x_0}],$$

and

$$[f][\bar{f}] = [e_{x_0}] = [\bar{f}][f].$$

Therefore

$$\pi_1(X, x_0) \text{ is a group,}$$

the **fundamental group of X** (at x_0).

For a noninteresting example, we have if $X \subseteq \mathbb{R}^n$ is convex, then

$$\pi_1(X, x_0) = 0$$

since straight line-homotopy contracts every loop to x_0. The interesting examples will be things like

$$S^1, \ \mathbb{R}^2\{0\}, \ T^2$$

where holes prevent certain contractions.

12.1 Base point is irrelevant

At first sight, $\pi_1(X, x_0)$ is dependent on the choice of x_0. Suppose $x_0, x_1 \in X$, and let

$$\alpha : I \to X$$

be a path from x_0 to x_1. Given a loop f based at x_0, we can transport the loop to x_1 by

$$\bar{\alpha} * f * \alpha,$$

where we should picture

$$x_1 \xrightarrow{\bar{\alpha}} x_0 \xrightarrow{f} x_0 \xrightarrow{\alpha} x_1.$$

Thus define

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$$

by

$$\hat{\alpha}([f]) = [\bar{\alpha} * f * \alpha],$$

getting an isomorphism. Thus if X is path-connected,

$$\pi_1(X, x_0) \cong \pi_1(X, x_1)$$

for any two basepoints.

Notice however that, while the groups are isomorphic, the isomorphism is dependent on the chosen path α. So it is in general sloppy to say "the fundamental group of X" without a basepoint, even if people often do so in the context of where the choice is harmless.

12.2 Simply connected space

Definition 12.1. A space is **simply connected** if

1. X is path connected, and
2. its fundamental group is trivial, i.e.

$$\pi_1(X, x_0) = 0.$$

Notice here that we don't care to talk about the base point since something like $\pi_1(X, x_0) = 0$ for some $x_0 \in X$ collapses to $\pi_1(X, x) = 0,\ \forall x \in X$ since the space is path connected.

Some nice examples are $\mathbb{R}^n$, convex sets, D^n.

However, S^1 is not simply connected.

A nice formulation used later will be X simply connected $\Rightarrow$ any two paths with the same endpoints are homotopic.

Intuitively, this should be true because if we suppose $f, g : x_0 \to x_1$, then

$$f * \bar{g}$$

is a loop at x_0. If X is simply connected,

$$[f * \bar{g}] = [e_{x_0}].$$

Algebraically cancelling g gives

$$[f] = [g].$$

12.3 Continuous maps induce homomorphisms

Suppose $h : (X, x_0) \to (Y, y_0)$ is continuous, meaning $h(x_0) = y_0$.

A loop $f : I \to X$ at x_0 gives a loop

$$h \circ f : I \to Y$$

at y_0. So define

$$h_* : \pi_1(X, x_0) \to \pi_1(Y, y_0)$$

by

$$h_*([f]) = [h \circ f].$$

This is well-defined as

$$f \simeq_p g \Rightarrow h \circ f \simeq_p h \circ g.$$

If

$$F : I \times I \to X$$

is the path homotopy, then

$$h \circ F : I \times I \to Y$$

is the required path homotopy.

Second,

$$h_*([f][g]) = h_*([f * g]) = [h \circ (f * g)].$$

But

$$h \circ (f * g) = (h \circ f) * (h \circ g),$$

so

$$h_*([f][g]) = h_*([f])h_*([g]).$$

Thus h_* is a group homomorphism.

12.4 Functoriality

If

$$X \xrightarrow{h} Y \xrightarrow{k} Z$$

then

$$(k \circ h)_* = k_* \circ h_*,$$

and

$$(\mathrm{id}_X)_* = \mathrm{id}_{\pi_1(X,x_0)}.$$

So we send

$$(X, x_0) \longrightarrow \pi_1(X, x_0)$$

preserving composition and identities. Smells **functorial**. In modern language,

$$\pi_1 : \mathbf{Top}_* \to \mathbf{Grp}$$

is a functor from **based** topological spaces to groups.

12.5 Fundamental group is a topological invariant

Suppose $h : X \to Y$ is a homeomorphism.

Then h^{-1} exists and is continuous. Functoriality gives

$$(h^{-1} \circ h)_* = (h^{-1})_* \circ h_* = (\mathrm{id}_X)_*.$$

Thus

$$(h^{-1})_* \circ h_* = \mathrm{id}\,.$$

Similarly

$$h_* \circ (h^{-1})_* = \mathrm{id}\,.$$

Hence

$$h_* : \pi_1(X, x_0) \xrightarrow{\sim} \pi_1(Y, h(x_0))$$

is an isomorphism.

Therefore

$$X \cong Y \Rightarrow \pi_1(X) \cong \pi_1(Y).$$

The contrapositive is especially useful,

$$\pi_1(X) \ncong \pi_1(Y) \Rightarrow X \ncong Y.$$

13 §53 Covering Spaces

At this point an issue arises. π_1 is elegant, but how do we calculate it? In contrast to homology groups, homotopy groups are in general uncomputable, even for arbitrary simplicial complexes.

13.1 Evenly covered neighborhoods

Let

$$p : E \to B$$

be continuous and surjective.

An open set $U \subseteq B$ is **evenly covered** by p if

$$p^{-1}(U)$$

can be written as a disjoint union of open sets

$$p^{-1}(U) = \bigsqcup_{\alpha} V_\alpha$$

such that each restriction

$$p\big|_{V_\alpha} : V \to U$$

is a homeomorphism.

We call the V_α's **slices** over U.

A map

$$p : E \to U$$

is a **covering map** if every $b \in B$ has some open neighborhood $U \ni b$ that is evenly

covered. Then E is called a **covering space** of B.

Locally, E looks like several disjoint copies of B. Over a sufficiently small U,

$$p^{-1}(U) = V_1 \sqcup V_2 \sqcup \cdots$$

and each

$$V_i \cong U.$$

So locally the picture is

$$\begin{aligned} V_3 &\cong U \\ V_2 &\cong U \\ V_1 &\cong U \\ &\downarrow p \\ &U \end{aligned}$$

13.2 The canonical example

Define

$$p : \mathbb{R} \to S^1$$

by

$$p(t) = (\cos 2\pi t, \sin 2\pi t).$$

This wraps the real line around the circle infinitely many times.

For example, take the open right semicircular region

$$U = \{(x, y) \in S^1 \mid x > 0\}.$$

The inverse image is

$$p^{-1}(U) = \bigsqcup_{n \in \mathbb{Z}} \left(n - \frac{1}{4}, n + \frac{1}{4}\right).$$

For every n,

$$p : \left(n - \frac{1}{4}, n + \frac{1}{4}\right) \longrightarrow U$$

is a homeomorphism. Therefore U is evenly covered.

Similar neighborhoods cover the whole circle, so

$$p : \mathbb{R} \to S^1$$

is a covering map.

13.3 WHy this should let us compute the fundamental group

The line $\mathbb{R}$ has no holes,

$$\pi_1(\mathbb{R}) = 0,$$

while the circle does. The covering map lets us study paths in S^1 by looking at corresponding paths upstairs in $\mathbb{R}$. The machinery for this, **lifting paths**, comes in §54.

So the eventual picture will be

- start a loop at $1 \in S^1$,
- lift it to a path in $\mathbb{R}$ starting at 0,
- because the original path ends where it began, the lifted path ends at some integer $n \in \mathbb{Z}$,
- that integer counts how many times the loop wound the circle,

yielding

$$\pi_1(S^1) \cong \mathbb{Z}.$$

13.4 Covering map vs. local homeomorphism

Every covering map is locally a homeomorphism.

Given $e \in E$, choose an evenly covered neighborhood U of $p(e)$. The point e lies in one slice V_α, and

$$p\,|_{V_\alpha} : V_\alpha \to U$$

is a homeomorphism.

So

$$\text{covering map} \implies \text{local homeomorphism}.$$

The converse, however, is false.

Munkres gives

$$p : \mathbb{R}^+ \to S^1$$

by

$$x \mapsto (\cos 2\pi x, \sin 2\pi x).$$

Locally, this is a homeomorphism, but near $(1, 0)$ the sheet at the endpoint of $\mathbb{R}^+$ is incomplete, so no neighborhood there is evenly covered.

13.5 Trivial coverings

If X is any space and D is discrete, consider

$$p : X \times D \to X, \;\; p(x, d) = x.$$

Then, for all open $U \subseteq X$,

$$p^{-1}(U) = \bigsqcup_{d \in D} U \times \{d\}.$$

Each slice maps homeomorphically to U.

Thus $X \times D$ is essentially a stack of separate copies of X.

13.6 Some structural facts

If $p : E \to B$ is a covering map, then p is an **open map**. Locally this is immediate since p is locally a homeomorphism.

Also if

$$B_0 \subseteq B$$

and

$$E_0 = p^{-1}(B_0),$$

then the restriction

$$p\big|_{E_0} : E_0 \to B_0$$

is again a covering map.

Products behave nicely,

$$p : E \to B,\ p' : E' \to B',$$

covering maps implies

$$p \times p' : E \times E' \to B \times B'$$

is a covering map.

Later this gives $\mathbb{R}^2 \to S^1 \times S^1$, the Torus.

13.7 Some problems

Problem 13.1. Let X be path connected and let $x_0, x_1 \in X$. Let α be a path from x_0 to x_1. Define

$$\hat{\alpha} : \pi_1(X, x_0) \to \pi_1(X, x_1)$$

by

$$\hat{\alpha}([f]) = [\bar{\alpha} * f * \alpha].$$

Prove that $\hat{\alpha}$ is an isomrophism.

Proof. We begin by showing it is a homomorphism.

Suppose $[f], [g] \in \pi_1(X, x_0)$. Then

$$\begin{aligned}\hat{\alpha}([f][g]) &= \hat{\alpha}([f * g]) \\ &= [\bar{\alpha} * (f * g) * \alpha] \\ &= [(\bar{\alpha} * f) * e_{x_0} * (g * \alpha)] \\ &= [(\bar{\alpha} * f) * \alpha * \bar{\alpha} * (g * \alpha)] \\ &= [(\bar{\alpha} * f * \alpha) * (\bar{\alpha} * g * \alpha)] \\ &= [\bar{\alpha} * f * \alpha][\bar{\alpha} * g * \alpha] \\ &= \hat{\alpha}([f])\hat{\alpha}([g]).\end{aligned}$$

Now for injectivity, suppose $\hat{\alpha}([f]) = \hat{\alpha}([g])$.

Then

$$\begin{aligned}[\bar{\alpha} * f * \alpha] &= [\bar{\alpha} * g * \alpha] \\ [\bar{\alpha} * f * \alpha] &= [\bar{\alpha}][g][\alpha] \\ \Rightarrow [\bar{\alpha} * f * \alpha][\bar{\alpha}][\bar{g}][\alpha] &= [e_{x_1}] \\ [\bar{\alpha} * f * \alpha * \bar{\alpha} * \bar{g} * \alpha] &= [e_{x_1}] \\ [\bar{\alpha} * f * \bar{g} * \alpha] &= [e_{x_1}] \\ [\alpha][\bar{\alpha} * f * g * \alpha][\bar{\alpha}] = [\alpha][e_{x_1}][\bar{\alpha}] &= [e_{x_0}] \\ [f][\bar{g}] &= [e_{x_0}] \\ \Rightarrow [f] &= [\bar{g}].\end{aligned}$$

For surjectivity, take some $[g] \in \pi_1(X, x_1)$. Then $[\alpha * g * \bar{\alpha}]$ is a loop at x_0 and thus in $\pi_1(X, x_0)$ with

$$\hat{\alpha}([\alpha * g\bar{\alpha}]) = [\bar{\alpha} * \alpha * g * \bar{\alpha} * \alpha] = [g].$$

Therefore $\hat{\alpha}$ is a group isomorphism. ■

Problem 13.2. Let

$$h : (X, x_0) \to (Y, y_0)$$

be continuous. Define

$$h_*([f]) = [h \circ f].$$

Prove from the definitions that h_* is well-defined and is a group homomorphism. Then prove

$$(h \circ k)_* = k_* \circ h_*$$

for a second based continuous map $k : Y \to Z$.

Proof. For $[f]$ we have that it is the class of all loops at x_0 that are homotopic.

So two f_1, f_2 are in $[f]$ means that there exists a continuous

$$\varphi : I \times I \to X$$

such that for every $s, t \in I$,

$$\varphi(s,0) = f_1(s),\ \varphi(s,1) = f_2(s),$$

and

$$\varphi(0,t) = x_0,\ \varphi(1,t) = x_0.$$

Consider

$$H = h \circ \varphi : I \times I \to Y.$$

It is continuous. Moreover,

$$H(s,0) = h(\varphi(s,0)) = h(f_1(s)) = (h \circ f_1)(s),$$

and similarly,

$$H(s,1) = (h \circ f_2)(s).$$

At the endpoints

$$H(0,t) = h(x_0) = y_0, \qquad H(1,t) = h(x_0) = y_0.$$

Therefore,

$$h \circ f_1 \simeq_p h \circ f_2.$$

Hence

$$[f_1] = [f_2] \Longrightarrow [h \circ f_1] = [h \circ f_2],$$

so

$$h_*([f]) = [h \circ f]$$

is well-defined.

For the homomorphism property take $[f], [g] \in \pi_1(X, x_0)$. Then

$$h_*([f][g]) = h_*([f * g]) = [h \circ (f * g)].$$

Use path concatenation to get

$$h \circ (f * g) = (h \circ f) * (h \circ g) = h_*([f])h_*([g]).$$

Finally, functoriality is immediate. For $[f] \in \pi_1(X, x_0)$,

$$\begin{aligned}(k \circ h)_*([f]) &= [(k \circ h) \circ f] \\ &= [k \circ (h \circ f)] \\ &= k_*([h \circ f]) \\ &= (k_* \circ h_*)([f]).\end{aligned}$$

■

Problem 13.3. Consider the covering map

$$p : \mathbb{R} \to S^1, \ p(t) = (\cos 2\pi t, \sin 2\pi t),$$

with basepoints $0 \in \mathbb{R}$ and $(1,0) \in S^1$.
A **lift** of a path $f : I \to S^1$ is a path

$$\widehat{f} : I \to \mathbb{R}$$

such that

$$p \circ \widehat{f} = f.$$

Compute $\pi_1(S^1)$.

Proof start. Given $\gamma : [0,1] \to S^1$ based at $(1,0)$, meaning $\gamma(1) = (1,0)$, there exists a unique lifted path

$$\hat{\gamma} : [0,1] \to \mathbb{R}$$

such that $\hat{\gamma}(0) = 0$ and $p \circ \hat{\gamma} = \gamma$.

SInce $\gamma(1) = (1,0)$, we have

$$p(\hat{\gamma}(1)) = (\cos(2\pi\hat{\gamma}(1)), \sin(2\pi\hat{\gamma}(1))) = (1,0).$$

The fiber over $(1,0)$ under p is $\mathbb{Z}$. Therefore, the endpoint $\hat{\gamma}(1)$ must be an integer

$$\hat{\gamma}(1) = n \in \mathbb{Z}.$$

We define a map

$$\phi : \pi_1(S^1, (1,0)) \to \mathbb{Z}, \ [\gamma] \mapsto \hat{\gamma}(1).$$

Suppose $\gamma_0 \simeq \gamma_1$ are path homotopic loops in S^1 relative to endpoints. Let $H : [0,1] \times [0,1] \to \mathbb{R}$ be a homotopy between the two loops. There exists a unique lifted homotopy $\hat{H}$ with codomain $\mathbb{R}$ such that

1. $\widehat{H}(s,0) = 0$ for all $s \in [0,1]$,
2. $p \circ \widehat{H} = H$.

Since $H(s,1) = (1,0)$ for all $s \in [0,1]$, the path $s \mapsto \hat{H}(s,1)$ lies entirely within the fiber $\mathbb{Z}$.

Because $[0,1]$ is connected and $\mathbb{Z}$ is discrete, the function $s \mapsto \hat{H}(s,1)$ must be constant. Thus

$$\hat{\gamma}_0(1) = \hat{H}(0,1) = \hat{H}(1,1) = \hat{\gamma}_1(1).$$

Thus ϕ is well-defined.

And so on... ■